

Generally Recognized as Safe (GRAS) Assessment for β -glucanase from *Talaromyces versatilis*

Prepared for:
Erbslöh Geisenheim GmbH

Prepared by:
Dicentra LLC
7 St. Thomas St., Suite 603, Toronto, ON
M5S 2B7

November 2024

Table of Contents

1	Part 1 – § 170.225 Signed Statements and Certification.....	5
1.1	GRAS Determination	6
1.2	Name and Address of Organization.....	6
1.3	Name of the Notified Substance.....	6
1.4	Intended Conditions of Use	6
1.5	Statutory Basis for GRAS Status.....	6
1.6	Premarket Exempt Status.....	6
1.7	Data Availability	6
1.8	Freedom of Information Act Statement.....	7
1.9	FSIS Statement	7
2	Part 2 – § 170.230 Identity, Method of Manufacture, Specifications and Physical or Technical Effect.....	7
2.1	Identity of Notified Substance	7
2.2	Name of GRAS Substance and Other Characteristics	7
2.2.1	Amino acid sequence and molecular mass	8
2.3	Production Microorganism	8
2.3.1	Production Strain	8
2.3.2	Antibiotic Resistance	9
2.3.3	Absence of Production Microorganism in Product	9
2.4	Description of Manufacturing Process	9
2.4.1	Manufacturing Site Information	9
2.4.2	Fermentation.....	9
2.4.3	Inoculum.....	10
2.4.4	Seed Fermentation.....	10
2.4.5	Main Fermentation	10
2.4.6	Recovery.....	11
2.4.7	Pre-Treatment	11
2.4.8	Primary Solid/Liquid Separation.....	11
2.4.9	Centrifugation	12
2.4.10	Filtration.....	12
2.4.11	Concentration	12
2.4.12	Polish and Germ Filtration	12

2.4.13	Formulation and Packaging	12
2.4.14	Immobilization Procedures.....	13
2.4.15	Manufacturing Process Flow Diagram	13
	A schematic overview of the manufacturing process for the Sumizyme PEG food enzyme is provided in Figure 1 below.....	13
2.4.16	Production Controls	15
2.5	Specifications and Batch Analysis	17
2.5.1	Specifications and Test Results for the Sumizyme PEG Food Enzyme	17
2.5.2	Specifications and Test Results for the Sumizyme PEG Food Enzyme Preparation.....	18
2.5.3	Stability of Sumizyme PEG.....	19
2.6	Enzymatic Activity	19
2.7	Technical Effect of Sumizyme PEG in Food	20
2.8	Use Levels	27
3	Part 3 – § 170.235 Dietary Exposure	28
4	Part 4 - § 170.240 Self-Limiting Use Levels	31
5	Part 5 - § 170. 245 Experience Based on Common Use in Food before 1958	31
6	Part 6 - § 170.250 GRAS Notice: Narrative	31
6.1	Safety of the production Strain.....	32
6.1.1	Pathogenicity and toxigenicity	33
6.2	Safety of the Enzyme Preparation	34
6.2.1	Safety of β -glucanase enzymes	34
6.2.2	Information on Side Activities of the Enzyme Which May Cause Adverse Effects	35
6.2.3	Allergenicity.....	35
6.2.4	Allergenicity Search.....	36
6.2.5	Toxicological testing.....	37
6.3	Safety Assessment and GRAS Determination.....	43
7	Part 7 - §170.255 List of Supporting Data and Information in GRAS	
	Dossier	44
7.1	Literature References.....	44
7.2	Appendix 1: Certificates of Analyses of the Food Enzyme and Food Enzyme Preparation	49

Table of Contents – Tables and Figures

Table 1. Compliance of Sumizyme PEG Food Enzyme with JECFA Specifications	17
Table 2. Analytical Test Results for 3 Batches of Sumizyme PEG Food Enzyme	18
Table 3. Specifications for Sumizyme PEG Enzyme Preparation	18
Table 4. Analytical Test Results for 3 Batches of the Sumizyme PEG Food Enzyme Preparation	19
Table 5. Recommended Use Levels of Sumizyme PEG.....	28
Table 6. Budget Method Assumed Daily Consumption of Food and Beverages	29
Table 7. Maximum Levels of Sumizyme PEG in Final Food Products.....	29
Table 8. Calculation of the Total TMDI	31
 Figure 1. Schematic Overview of the Manufacturing Process of Sumizyme PEG Food Enzyme from <i>Talaromyces versatilis</i>	 14

Generally Recognized as Safe (GRAS) Assessment for β -glucanase from *Talaromyces versatilis*

1 Part 1 – § 170.225 Signed Statements and Certification

In accordance with 21 CFR 170 Subpart E, Erbslöh Geisenheim GmbH (herein referred to as “Erbslöh”) hereby states that β -glucanase from *Talaromyces versatilis*, manufactured by Erbslöh, is not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act (FFDCA) based on Erbslöh’s review that the product is Generally Recognized as Safe (GRAS) under the conditions of its intended use described in Section 1.4 Intended Conditions of Use below. In addition, as the responsible official of Erbslöh, Dr. Jürgen Meinel, hereby certifies that all data and information presented in this dossier is complete, representative, and balanced, and which considered all unfavorable as well as favorable information known to Erbslöh and pertinent to the evaluation of the safety and GRAS status of the intended use of β -glucanase from *Talaromyces versatilis* as an ingredient for addition to food.



Dr. Jürgen Meinel
Head of Quality & Regulatory

Date: November 15, 2024

1.1 GRAS Determination

Erbslöh asserts that this GRAS conclusion for their ingredient “ β -glucanase from *Talaromyces versatilis*” has been reached in accordance with the requirements of 21 CFR 170, Subpart E.

1.2 Name and Address of Organization

Erbslöh GmbH
Erbslöhstraße 1
D-65366 Geisenheim
Germany

1.3 Name of the Notified Substance

The name of the substance of this GRAS assessment is β -glucanase from *Talaromyces versatilis*. Its trade name is Sumizyme PEG, for which it will be referred to as herein.

1.4 Intended Conditions of Use

Sumizyme PEG is intended to be used in cereals for brewing and grapes for wine making, as well as in the processing of fruits, vegetables and yeast. The enzyme is used at levels no higher than the necessary dosage to achieve the desired enzymatic reaction and according to Good Manufacturing Practice. Consequently, there are no maximal use levels, only recommended dosages.

1.5 Statutory Basis for GRAS Status

Sumizyme PEG has been determined to be GRAS through scientific procedures in accordance with 21 CFR Part 170.30(a) and (b) of the *Code of Federal Regulations* (CFR) for use as a food ingredient in certain specific categories of food.

1.6 Premarket Exempt Status

Sumizyme PEG is not subject to the premarket approval requirements of the Federal Food, Drug and Cosmetic Act based on the conclusion that the substance in question has been determined to be GRAS under the conditions of its intended use.

1.7 Data Availability

The data and information that has been used to prepare the rationale that concludes the ingredient Sumizyme PEG to be GRAS under the conditions of its intended use are provided within this notice. Any additional information will be made available upon request by the United States

Food and Drug Administration (FDA), or will be available for review and copying at the offices of Erbslöh.

Erbslöh GmbH
Erbslöhstraße 1
D-65366 Geisenheim
Germany

1.8 Freedom of Information Act Statement

None of the information found within this GRAS dossier is exempt from disclosure under the Freedom of Information ACT, 5 U.S.C. 552, except for confidential manufacturing information.

1.9 FSIS Statement

Erbslöh does not intend to add Sumizyme PEG to any meat and/or poultry products that come under USDA jurisdiction. Therefore, 21 CFR 170.270 does not apply.

2 Part 2 – § 170.230 Identity, Method of Manufacture, Specifications and Physical or Technical Effect

2.1 Identity of Notified Substance

This dossier concerns an endo-1,3(4)- β -glucanase enzyme preparation from *Talaromyces versatilis*. The trade name ‘Sumizyme PEG’ will be used to describe this endo-1,3(4)- β -glucanase enzyme throughout the dossier.

2.2 Name of GRAS Substance and Other Characteristics

Name of Enzyme: endo-1,3(4)- β -glucanase/beta-glucanase

Synonyms: endo-1,3- β -D-glucanase; laminarinase; laminaranase; β -1,3-glucanase; β -1,3-1,4-glucanase; endo-1,3- β -glucanase; endo- β -1,3(4)-glucanase; endo- β -1,3-1,4-glucanase; endo- β -(1 $\rightarrow$ 3)-Dglucanase; endo-1,3-1,4- β -D-glucanase; endo- β -(1-3)-D-glucanase; endo- β -1,3 glucanase IV; endo-1,3- β -D-glucanase; 1,3-(1,3;1,4)- β -D-glucan 3(4)-glucanohydrolase

Trade Names: Sumizyme PEG

EC (IUBMB) number: EC 3.2.1.6

CAS number: 62213-14-3

EINECS number: 263-462-4

Production strain: *Talaromyces versatilis* strain PF8 (formerly *Penicillium funiculosum*)

2.2.1 Amino acid sequence and molecular mass

Based on an SDS-PAGE (sodium dodecyl sulfate–polyacrylamide gel electrophoresis) analysis, it was determined that the molecular mass of the enzyme protein is 68 kDa, while the number of amino acids within the β -glucanase protein has been determined to be 679.

2.3 Production Microorganism

2.3.1 Production Strain

The source material of Sumizyme PEG is the fungus *Talaromyces versatilis* sp. nov. IF550186 (Bridge, 2013). The former systematic name *Penicillium funiculosum* has been recently replaced by *Talaromyces versatilis* (Samson *et al.* 2011).

The identification of the strain was confirmed by TechnoSuruga Laboratory Co., Ltd., based on internal Transcribed Spacer 1 and 2 (ITS1 and ITS2) sequences and the 5.8S gene.

Talaromyces versatilis is a type of filamentous fungus. Its colonies exhibit a characteristic pattern with a tendency to form aerial ropes (funiculose hyphae) interspersed with wooly (floccose) regions. The reverse coloration typically ranges from yellow to orange or red, and on malt agar, it demonstrates good sporulation. Conidiophores emerge perpendicular to the funiculose hyphae, with those in marginal regions originating directly from the substrate hyphae. They terminate in biverticillate, symmetric penicilli carrying conidial heads that are dark green to bluish in color. The conidia, as well as the conidiophores, are smooth. The optimal growth temperature for *Talaromyces versatilis* is 20-40 °C. It is capable of growing at a wide pH range, though acidic conditions are more favourable. *Talaromyces versatilis* favours high oxygen concentrations and therefore grows best on the surface of organic substrates. It uses asexual reproduction via formation of conidiospores (Singh *et al.* 1986; van Reenen-Hoekstra *et al.* 1990).

The organism has not been genetically modified to produce the enzyme. The species *Talaromyces versatilis* has been used for decades in the production of food and feed enzymes, under its former systematic name *Penicillium funiculosum*. *Talaromyces versatilis* has a long history of safe use in industrial-scale enzyme production, including a previous GRAS determination for a cellulase enzyme from *Penicillium funiculosum* (GRN 584), which received no questions from the FDA. More recently, the European Food Safety Authority (EFSA) authorized the use of the same enzyme that this the subject of this GRAS notice; endo-1,3(4)- β -glucanase produced from *Talaromyces versatilis* strain PF8 (EFSA, 2024). Thus, *Talaromyces versatilis* is recognized as a safe production organism for enzymes for food as well as feed processing applications.

The taxonomic classification of this microorganism is as follows:

Kingdom: Fungi
Division: *Ascomycota*
Class: *Eurotiomycetes*
Order: *Eurotiales*
Family: *Trichocomaceae*
Genus: *Talaromyces*
Species: *Talaromyces versatilis*
Strain: *Talaromyces versatilis* PF8

2.3.2 Antibiotic Resistance

No antibiotic resistance genes were introduced in the production microorganism, and antibiotics are not produced by the source organism.

2.3.3 Absence of Production Microorganism in Product

Each batch of the enzyme preparation is strained through a ceramic filter with a pore size of 0.2 µm. Thus, microorganisms larger than 0.2 µm completely removed to ensure the absence of the production strain, *Talaromyces versatilis* in the final product. Strict control of all manufacturing steps is applied under the certified quality control system.

2.4 Description of Manufacturing Process

2.4.1 Manufacturing Site Information

The food enzyme is produced by Shin Nihon in accordance with current Good Manufacturing Practices for Food (GMP) and the principles of Hazard Analysis of Critical Control Points (HACCP).

2.4.2 Fermentation

The fermentation process to produce food enzymes from microbial sources is described below. Fermentation commonly occurs in many different food products, and this fermentation process has been used in the production of food enzymes for decades,

The fermentation processes used to produce Sumizyme PEG is a solid-state fermentation, with the following main steps:

- Inoculum
- Seed fermentation
- Main fermentation

The raw materials used during the fermentation and recovery stages of the manufacturing process are of food grade quality. Furthermore, all raw materials used for the production of Sumizyme PEG are food, food additives, or food processing aids listed in the Food Chemical Codex (13th edition) and/or the Code of Federal Regulations (“21 CFR”), and comply with their respective specifications. Any antifoams used in the fermentation and recovery are used in accordance with the Enzyme Technical Association submission to FDA on antifoams and flocculants in April, 1998.

Materials used in the fermentation process (inoculum, seed and main fermentation) are:

- Potable water
- A carbon source
- A nitrogen source
- Vitamins
- Salts and minerals
- pH adjustment agents
- Foam control agents

2.4.3 Inoculum

A pure culture suspension of *Talaromyces versatilis* is transferred under sterile conditions to fermentation medium in an inoculum flask. The culture is grown in the flask until a sufficient amount of biomass is obtained, which can subsequently be used as inoculum for the seed fermentation.

2.4.4 Seed Fermentation

Under sterile conditions, the inoculum is then transferred fermentation medium within the seed fermentor. The seed fermentation is run under the following process conditions:

- Temperature: 30 °C
- For 1 to 3 days
- pH : 4.5 to 5.5

The fermentation is run under controlled agitation and controlled air supply. When a sufficient amount of biomass has developed, the content of the seed fermenter is used for inoculation of the main fermentation.

2.4.5 Main Fermentation

During the main fermentation, the production organism biosynthesizes the enzyme protein. The fermentation in the main fermenter is operated as a batch process. The content of the seed fermenter is mixed with the solid fermentation medium and subsequently transferred aseptically onto fermentation trays or into a deep bed fermenter. In both cases, the fermentation is run under a controlled atmosphere. Where necessary, agitation is applied in order to provide for suitable

oxygen supply in the whole fermenter. The solid state fermentation is run under the following process conditions:

- Temperature: 30 °C
- Humidity 90 - 100 %

The fermentation process proceeds either for a set duration, until laboratory analyses confirm the desired enzyme production has been achieved, or until the rate of enzyme production decreases below a predetermined threshold. When either of these conditions are met, typically after 4 days, the fermentation is completed.

2.4.6 Recovery

The purpose of the recovery process is:

- to isolate the desired enzyme protein from the fermentation substrate and biomass,
- to concentrate the desired enzyme protein and to enhance the enzyme activity/Total Organic Solids (TOS) ratio

Throughout fermentation, the enzyme protein is secreted by the microorganism *Talaromyces versatilis* into the fermentation medium. During the recovery process, the enzyme is isolated from the biomass. Depending on the specific enzyme production plant, the nature, amount and order of the different types of unit operations described below may vary:

- Pre-treatment
- Primary solid/ liquid separation
- Concentration
- Polish and germ filtration

Materials used during recovery of the food enzyme, if necessary, include filter aids, potable water and pH adjustment agents.

2.4.7 Pre-Treatment

Depending on the process used for subsequent solid separation, water and diatomaceous earth are added to the fermentation substrate.

2.4.8 Primary Solid/Liquid Separation

The purpose of the primary separation is to remove the biomass solids from the enzyme protein. The solids consist of the biomass and fermentation substrate. During the primary separation, the solid fermentation substrate is submerged in an aqueous extraction medium to separate the enzyme protein from the fermentation substrate. It is performed at defined pH ranges (4 to 4.5) in order to minimize the loss of enzyme activity. To achieve this, one or more of the following unit operations are used:

2.4.9 Centrifugation

Centrifugation is applied to accelerate the natural sedimentation of the solids in the aqueous extraction medium. The liquid (centrifugate) containing the enzyme protein is collected for further recovery and formulation. The centrifugation equipment is tubular bowl centrifuges: Use high centrifugal force and are used batch-wise to sediment particles.

2.4.10 Filtration

Filtration is a technique used to separate the aqueous extraction medium into a solid and a liquid (filtrate) fraction by forcing the broth over a filter. This retains the solids and allows the liquid to pass through. The filtrate containing the enzyme protein is collected for further recovery and formulation. Here microfiltration is used, based on a continuous cross flow membrane filtration process. The extraction medium is pumped over the filter membrane, during which the enzyme-containing liquid (permeate) passes through the membrane pores, whereas the solids stream (retentate) is pumped away.

2.4.11 Concentration

The solution containing the enzyme protein must undergo concentration to achieve the intended enzyme activity and/or to increase the enzyme activity/TOS ratio prior to formulation. Temperature and pH are regulated during this concentration step, and continues until the desired concentration is achieved. Ultrafiltration is used in that step, which is a membrane filtration process applied to increase the ratio of high molecular weight molecules (such as the enzyme protein) versus the low molecular weight molecules. It increases the activity/TOS ratio.

2.4.12 Polish and Germ Filtration

For removal of residual solids from the fermentation and as a general precaution against microbial contamination, filtration on dedicated germ filters is applied at various stages during the recovery process. If required, pre-filtration (polish filtration) is integrated to eliminate insoluble substances and facilitate the germ filtration using diatomaceous earth and filter press. At the end of the recovery process, the final polish- and germ filtration (centrifugation and 0.2 µm ceramic filtration) produces a concentrated enzyme solution free of insoluble substances and the production strain. The enzyme solution before formulation is defined as “food enzyme.”

2.4.13 Formulation and Packaging

Subsequently, the food enzyme is formulated, and the resulting product is defined as a “food enzyme preparation.” Food enzymes can be sold as dry or liquid preparations, depending on the final application where the enzyme is intended to be used. For the manufacture of liquid food enzyme preparations, the food enzyme is standardized with food grade ingredients and adjusted to the desired activity. The preparation is stabilized with preservatives in certain cases.

The food enzyme preparation is tested for quality control of all relevant parameters, such as enzyme activity and the general Joint FAO/WHO Expert Committee on Food Additives (JECFA) Specifications for Food Enzyme Preparations. Prior to storage, the final product is packaged in suitable food packaging material. Warehousing and transportation conditions are specified on the respective product label for food enzyme preparations.

The nutrient consumed by the microorganism in the fermentation process is wheat bran. No other major allergen substances will be used in the fermentation and recovery processes.

2.4.14 Immobilization Procedures

The food enzyme is not immobilized.

2.4.15 Manufacturing Process Flow Diagram

A schematic overview of the manufacturing process for the Sumizyme PEG food enzyme is provided in Figure 1 below.

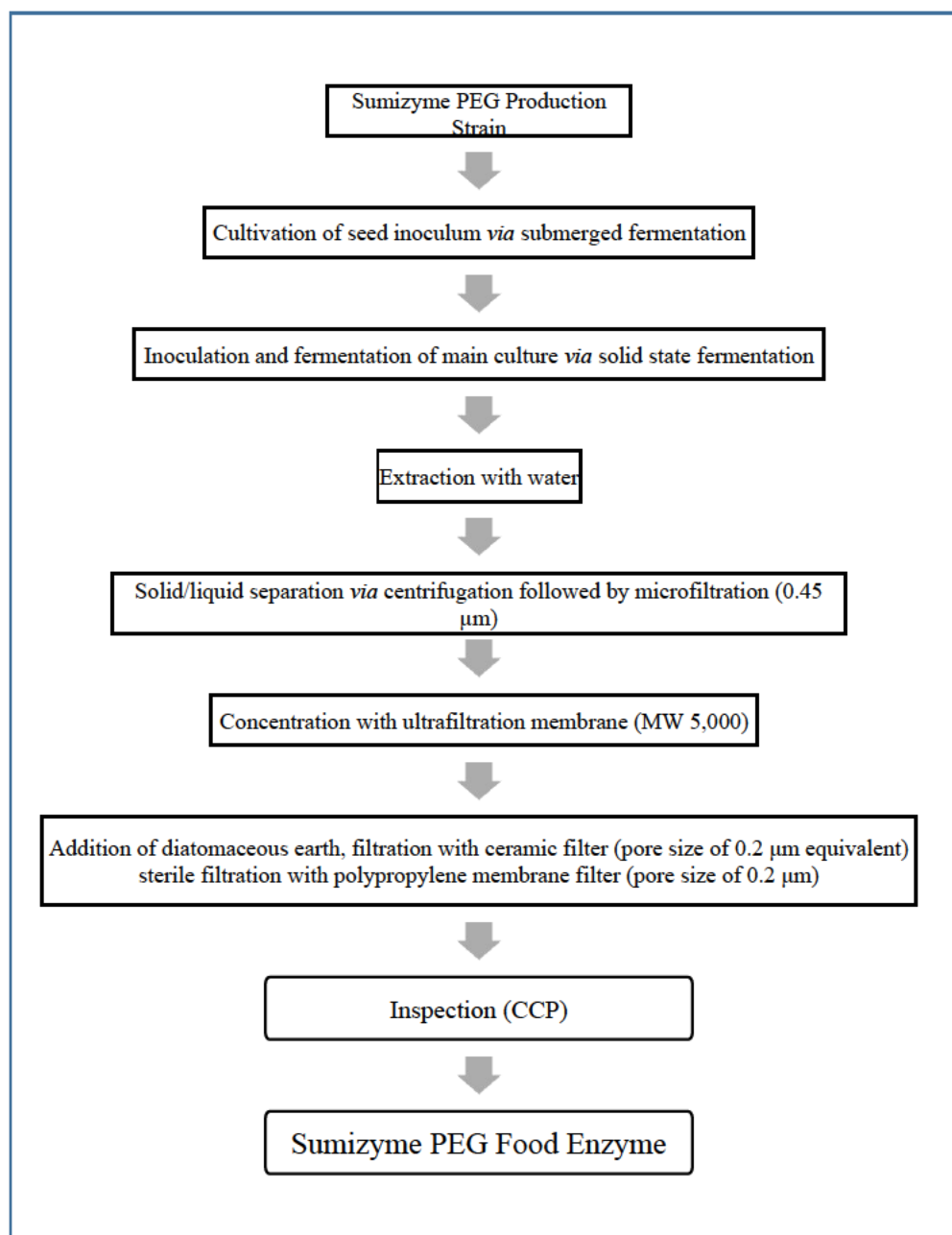


Figure 2. Schematic Overview of the Manufacturing Process of Sumizyme PEG Food Enzyme from *Talaromyces versatilis*

2.4.16 Production Controls

Good Manufacturing Practice and HACCP

Potential Hazards

To comply with current GMP and HACCP principles for food manufacturing, the following potential hazards associated with the production of the food enzyme are considered and controlled throughout the production process:

Identity and Purity of the Producing Microorganism

It is of great importance to the manufacturer of the food enzyme to ensure that the production microorganism effectively produces the desired enzyme protein. Hence, it is crucial to control the identity and purity of the microorganism.

Microbiological Hygiene

For optimal production of the enzyme, it is critical to maintain hygienic conditions throughout the whole fermentation process. Any microbial contamination would promptly hinder the growth of the production organism, leading to reduced yield of the desired enzyme protein and potentially resulting in product rejection.

Chemical Contaminants

It is also of importance that the raw materials used during the fermentation steps are of suitable quality and are free of contaminants which may jeopardize the safety of the food enzyme and/or impede the optimal growth of the production organism, which may consequently affect enzyme yield.

Control Measures

The main measures to control the hazards identified above are:

Purity of the Producing Microorganism

The production of the required enzyme protein relies on a well-defined Master (MCB) and Working Cell Bank (WCB). A Cell Bank consists of ampoules containing a pure culture, with the cell line's history and the processes involved in producing, propagating, preserving, and storing it being closely monitored and regulated.

The MCB is established from a selected strain, while the WCB is produced by sub-culturing of one or more ampoules from the MCB. The acceptance of a WCB for production runs depends on meeting specific standards, determined by assessing its identity, viability, microbial purity, and productivity. An approved WCB serves as the seed material for the inoculum.

Microbiological Hygiene

Measures to guarantee microbiological hygiene and prevent contamination with microorganisms pervasively present in the environment (air, water, raw materials) are:

- Hygienic design of equipment
Cleaning and sterilization:
 - Validated standard sterilization and cleaning procedures of the equipment and process workspace
 - Sterilization of all fermentation media
 - Usage of sterile air for fermentor aeration
- Hygienic processing:
 - Contents of the WCB ampoule, inoculum flask or seed fermenter are aseptically transferred.
 - Positive pressure maintenance in the fermenter
- Germ filtration

Chemical Contaminants

It is ensured that all raw materials used in the food enzyme production are of food grade quality or have been evaluated to be suitable for their intended purpose and comply with agreed specifications.

In-Process Testing and Monitoring

In-process testing and monitoring is performed in addition to the control measures described above, to ensure a high-quality product and an optimal and effective enzyme production process (GMP).

These in-process controls comprise:

Microbial Controls

Absence of significant microbial contamination is analysed by microscopy or plate counts before inoculation of both the seed and main fermentation and at regular intervals, and at critical process steps during fermentation and recovery.

Monitoring of Fermentation Parameters

Monitoring of fermentation parameters include:

- Temperature
- Oxygen content
- CO₂ content

Throughout the fermentation process, the values of the parameters above are constantly monitored.

These values serve as indicators of whether sufficient biomass or enzyme protein has been produced and the fermentation process progresses according to plan. Any deviations from the predetermined values prompt adjustments to ensure an optimal and robust process.

Enzyme activity is also monitored at regular intervals and at critical steps throughout the entire food enzyme production process.

2.5 Specifications and Batch Analysis

2.5.1 Specifications and Test Results for the Sumizyme PEG Food Enzyme

It is proposed that the food enzyme should comply with the globally recognized JECFA specifications for the microbiological and chemical purity of food enzymes (FAO/WHO, 2006):

Lead: Not more than 5 mg/kg

Salmonella sp.: Absent in 25 g of sample

Total coliforms: Not more than 30 per gram

Escherichia coli: Absent in 25 g of sample

Antimicrobial activity: Not detected

The Sumizyme PEG Food enzyme complies with these specifications as demonstrated by the analyses of 3 batches in Table 1 below. Additional analytical test results for 3 batches of the Sumizyme PEG food enzyme are provided in Table 2 below (See Appendix 1 for Certificates of Analyses). All analytical methods used for each parameter on the batches of Sumizyme PEG provided have been validated for their respective intended uses.

Table 1. Compliance of Sumizyme PEG Food Enzyme with JECFA Specifications

Test	Batch Analysis		
	110402T	120402T	140303T
Lead (mg/kg)	<0.05	<0.05	<0.05
<i>Salmonella</i> sp. (per 25 g)	Not detected	Not detected	Not detected
Total coliforms (per g)	<30	<30	<30
<i>Escherichia coli</i> (per 25 g)	Not detected	Not detected	Not detected
Antimicrobial activity	Not detected	Not detected	Not detected

Lead: Atomic Absorption Spectrophotometry

Salmonella: AOAC Official Methods of Analysis (1990)

E. Coli, Total Coliforms: FDA bacteriological Analytical manual 7th edition (1992)

Antimicrobial activity: Specifications for Identity and Purity of Certain food Additives, FAO Food and Nutrition Paper 49

Table 2. Analytical Test Results for 3 Batches of Sumizyme PEG Food Enzyme

Test	Test Method	Batch Analysis		
		110402T	120402T	140303T
Appearance	Organoleptic and Microscopic evaluation	Brown Liquid	Brown Liquid	Brown Liquid
Ash (%)	JECFA. 1st to the 65th meeting (1956-2005). Volume 4. 2006. ISBN 9789251055694.	0.8	0.8	0.9
Water (%)	JECFA. 1st to the 65th meeting (1956-2005). Volume 4. 2006. ISBN 9789251055694.	78.6	78.2	77.9
Protein (%)	JECFA. 1st to the 65th meeting (1956-2005). Volume 4. 2006. ISBN 9789251055694.	10.8	10.9	11.1
TOS (%)	Dry Matter (%) minus Ash (%)	20.6	21.0	21.2
β -glucanase activity (units/g)	OIV/OENO 340-2010 EN	827	883	830
Units/mg TOS	(β -glucanase activity)/mg TOS	4.0	4.2	3.9

JECFA= Joint FAO/WHO Expert Committee on Food Additives; OIV = International Organization of Vine and Wine; TOS = Total Organic Solids

2.5.2 Specifications and Test Results for the Sumizyme PEG Food Enzyme Preparation

The specifications and analytical results for 3 batches of the Sumizyme PEG food enzyme preparation are provided below in Tables 3 and 4, respectively.

Table 3. Specifications for Sumizyme PEG Enzyme Preparation

Parameters	Specifications	Test Methods
Chemical Contaminants		
Heavy Metals		
Arsenic (mg/kg)	< 0.1	DIN EN ISO 15763 (2010), CON-PV 01274 (2022-06), ICP-MS
Lead (mg/kg)	< 0.05	
Cadmium (mg/kg)	< 0.01	
Mercury (mg/kg)	< 0.005	
Microbiological Contaminants		
Coliforms (CFU/g)	< 10	§ 64 LFGB L01.00-03, 1987-03; VA10-009-V07, Kulturelles Zählverfahren (nicht-chromogene Nährböden)
<i>E. Coli</i> (per g)	Not detected	Eurofins Internal Method, 2002-12; VA10-010-V09, Cultural detection (non-chromogenic culture media)
<i>Salmonella</i> (per 25 g)	Not detected	AFNOR EGS 38/01-03/15, 2015-01; VA10-050-V10, Real-time PCR
<i>S. Aureus</i> (CFU/g)	< 10	ISO 6888-1, 2019-06; VA10-007-V09, Kulturelles Zählverfahren (nicht-chromogene Nährböden)

DIN = Deutsches Institute für Normung e.V (German Institute for Standardization); ISO = International Standards Organization; LFGB = Lebensmittel-, Bedarfsgegenstände-und Futtermittelgesetzbuch (German Food and Feed Code)

Table 4. Analytical Test Results for 3 Batches of the Sumizyme PEG Food Enzyme Preparation

Test	Specifications	Batch Analysis		
		209-2023-00022320	209-2023-00022321	209-2023-00022322
Heavy Metals				
Arsenic (As)	< 0.1	< 0.1	< 0.1	< 0.1
Lead (Pb)	< 0.05	< 0.05	< 0.05	< 0.05
Cadmium (Cd)	< 0.01	< 0.01	< 0.01	< 0.01
Mercury (Hg)	< 0.005	< 0.005	< 0.005	< 0.005
Microbiological Contaminants				
Coliforms (CFU/g)	< 10	< 10	< 10	< 10
<i>E. Coli</i> (per g)	Not detected	Not detected	Not detected	Not detected
<i>Salmonella</i> (per 25 g)	Not detected	Not detected	Not detected	Not detected
<i>S. Aureus</i> (CFU/g)	< 10	< 10	< 10	< 10

For the manufacture of the liquid food enzyme preparation, the food enzyme is combined with food grade ingredients listed below, and adjusted to the to the desired enzyme activity. A variety of commercial formulations exist, while the following is a representative composition:

Enzymatic Activity: 300 U/g
TOS: 7.3%
Glycerol: 53%
Water: 39.5%
pH: 4 - 4.5

2.5.3 Stability of Sumizyme PEG

Food enzymes are not sold as such, but are formulated into different enzyme preparations to achieve stable and standardized products. The stability of the final product thus depends on the type of formulation, and not just on the food enzyme itself. The product labels of the food enzyme preparation specify the date of minimum durability or use-by-date, and may also include specific storage conditions, if required.

2.6 Enzymatic Activity

Beta-glucanase hydrolyses β -D-glucans with (1 $\rightarrow$ 3)-linkages and/or (1 $\rightarrow$ 4)-linkages. Beta-glucanases commonly use a broad range of substrates. In the present case, substrates include cereal D-glucans, laminarin, lichenin, yeast glucan and other fungal glucans (Kikuchi *et al.* 2005, Hua *et al.* 2011, Hong *et al.* 2002). This substrate versatility demonstrates the usefulness of β -glucanase in various food processing applications, where related β -glucans are present.

The enzyme activity data shown in section 2.5 were obtained by an analytical method provided International Organisation of Vine and Wine (OIV/OENO 340-2010 EN), with enzyme activity reported in β -glucanase Units/g.

The enzyme activity of β -glucanases has also been identified in many other sources, including plants, microorganisms and animals, while β -glucanase as a food enzyme has a history of several decades of safe use as food processing agent (Pariza and Foster, 1983).

2.7 Technical Effect of Sumizyme PEG in Food

In principle, the enzymatic conversion of β -glucan catalyzed by β -glucanase is beneficial in the processing of all food raw materials which naturally contain this substrate.

The benefits of the use of industrial β -glucanase in typical food or food ingredient processes are described below for various applications. Food producers find these effects valuable as they enhance processability and/or product quality. Additionally, these applications contribute to more efficient production processes, thereby enhancing production efficiency and providing environmental advantages such as reduced raw material usage and waste generation.

These benefits in certain food processes include:

Brewing:

- Reduction of viscosity, increase of filterability.
- Increase of yield/extraction from malt.
- Better production economy and environmental benefits, such as the use of less raw materials, energy savings, and production of less waste.

Fruit and vegetable processing:

- Increased compound extraction (colour, flavour, or phenolics).
- Increased clarification or filterability.
- Better production economy and environmental benefits, such as the use of less raw materials, energy savings, and production of less waste.

Wine:

- Reduction of clarification time.
- Reduction of viscosity, increase of filterability.
- Increased wine yeast lysis.
- Better production economy and environmental benefits, such as the use of less raw materials, energy savings, and production of less waste.

Yeast extract processing:

- Increased lysis of yeast cell walls.
- Increase extraction / yields of yeast extracts.

- Better production economy and environmental benefits, such as the use of less raw materials, energy savings, and production of less waste.

Effect of the Presence of (Residues of) the Food Enzyme on the Final Food

It is not the food enzyme itself, but the result of the enzymatic conversion that determines the effect in the food or food ingredient (including raw materials). This effect remains, irrespective of whether the food enzyme is still present or removed from the final food.

The effect of β -glucanase is the cleavage of β -glucans in cereals, fruits, vegetables and yeasts during food processing, resulting in glucan-oligomers and glucose. The enzyme does not exert any (unintentional) enzymatic activity in the final food, due to a combination of various factors, depending on the application and the process conditions used by the individual food producer. These factors include depletion of the substrate, denaturation of the enzyme during processing, lack of water activity, insufficient pH, etc. Consequently, the presence of residues of the food enzyme in the final food does not lead to an effect in or on the final food.

Effect of the use of the Food Enzyme on the Final Food

The majority of food enzymes perform their technological function during food processing. In many cases, they are used merely to improve the production process of a certain food or food ingredient, and the impact of the enzymatic conversion is not noticeable in the final food product. In other cases, however, the effect of the enzymatic conversion (depletion of a substrate and/or development of a reaction product) might also result in a beneficial effect on the final food. Below, the beneficial effects that can be obtained in the final foods when produced with the help of β -glucanase are described.

Brewing:

By application of β -glucanase in the brewing process, the following characteristics of the final product beer can be achieved:

- Prevention of β -glucan haze formation in the final product

Fruit and Vegetable Processing:

Due to better processing properties (e.g. reduced viscosity for better solid-liquid separation) the beneficial effects are:

- Improved extraction of valuable substances from the fruits and vegetables in the juice (e.g. enhanced nutrients levels and colour)
- Improving appearance and taste of juices, meeting consumers' expectations (by e.g. reducing post-processing hazes (i.e. clearer product))

Wine Making:

Processing improvement (lowering viscosity, improving filterability and wine clarification) leads to the following beneficial effects on the final food:

- Improved wine organoleptic properties (mouth feeling and aroma by better extraction of aroma precursors from yeast)

Yeast Extract Processing:

Improved yeast lysis leads to the following beneficial effects on the final food:

- Improved organoleptic and nutritional properties of yeast extracts (better extraction of aroma components from yeast)

Fate of the Enzyme in Food

β -Glucanase performs its technological function during food processing. The food enzyme has been used for many decades as a processing aid in food production (Pariza and Foster, 1983). The enzyme catalyses the hydrolysis of the β -1,3- and / or β - 1,4-linkages in β -D-glucans like cereal or fungal β -D-glucans. Like the endogenous β -glucanase present in the raw materials and ingredients of common foods, it does not perform any technological function in the final food product.

Based on the conditions of use described in Section 1.4 and the activity of β -glucanase under such conditions, it can be concluded that the β -glucanase enzyme does not exert any (unintentional) enzymatic activity in the final food.

Fate of the Enzyme Protein During Brewing

Beer is pasteurized before going onto the market to ensure the microbiological stability of the product and are therefore denatured during the brewing process.

β -Glucanases are typically added in the mashing step or to the adjunct before addition of the adjunct to the mash tun, and are therefore already denatured in the consecutive lautering or mash filtration step. β -Glucanases can also be added during the fermentation step, for which in this case, β -glucanases will be denatured during the pasteurization step.

Fate of the Enzyme Protein During Fruit and Vegetable Processing

Fruits and vegetables products, such as juices, jam, fruit preparations and canned fruits, are pasteurized before going onto the market to ensure the microbiological stability of the product. β -Glucanases are therefore denatured during fruit and vegetable processing.

Fate of the Enzyme Protein During Wine Making

In the wine making process, the β -glucanase is added before clarification- or filtration steps. At the end of the wine making process, one or more of the following unit operations are to be used:

- Bentonite is added into white and rosé wines prior to filtration (and bottling) in order to stabilize the wine by the removal of thermo-labile proteins. Enzymes present in the wine are absorbed and therefore removed with the help of bentonite addition.

- Wine is filtrated on membranes that remove enzyme protein (the cut-off of the membrane - usually 20 kDa- is smaller than the molecular size of the enzyme).
- In certain rare cases, wines are heat-treated.
- Natural wine ingredients such as alcohol, polyphenols, heavy metals, sulphur in form of SO₂, inhibit and naturally precipitate the enzyme (forming the so-called tannin protein cloudiness).

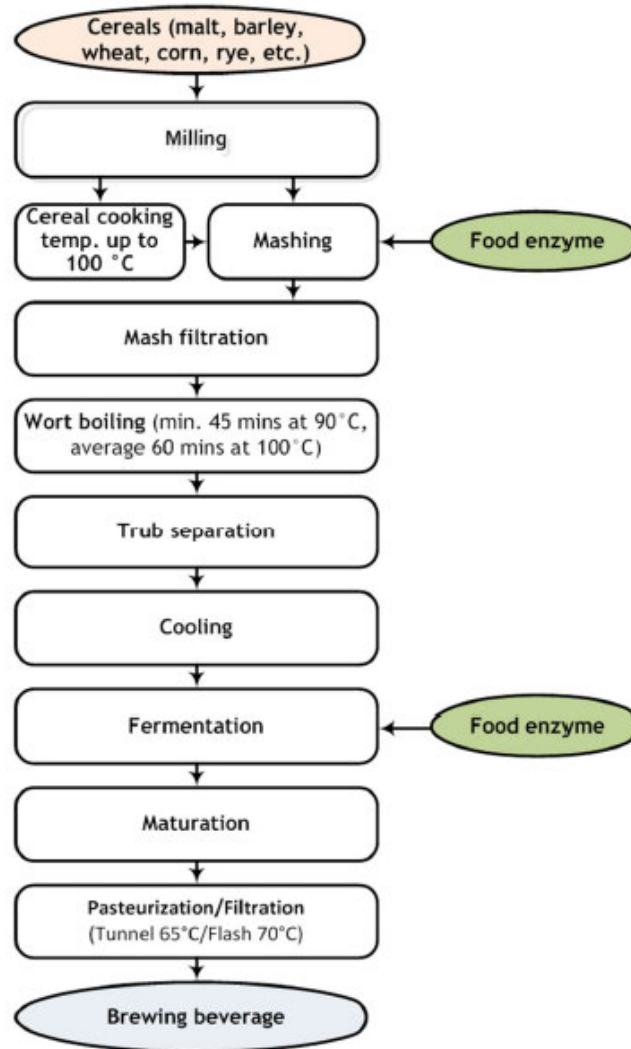
Fate of the Enzyme Protein During Yeast Processing

The β -glucanase can be added at different steps of the yeast processing. The enzyme proteins are denatured by ultra high temperature (UHT) treatment or by other heating processes during the evaporation or drying step.

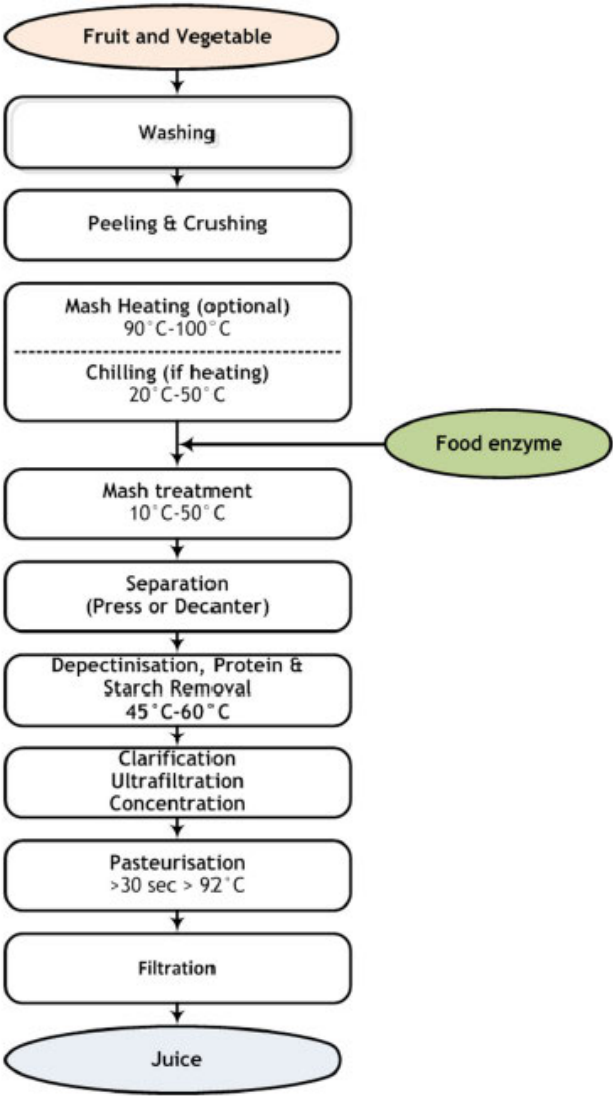
Food Production Process Flow Charts

The food production process flow charts provided below comprise typical applications of the food enzyme and show the conditions under which the food enzyme is used.

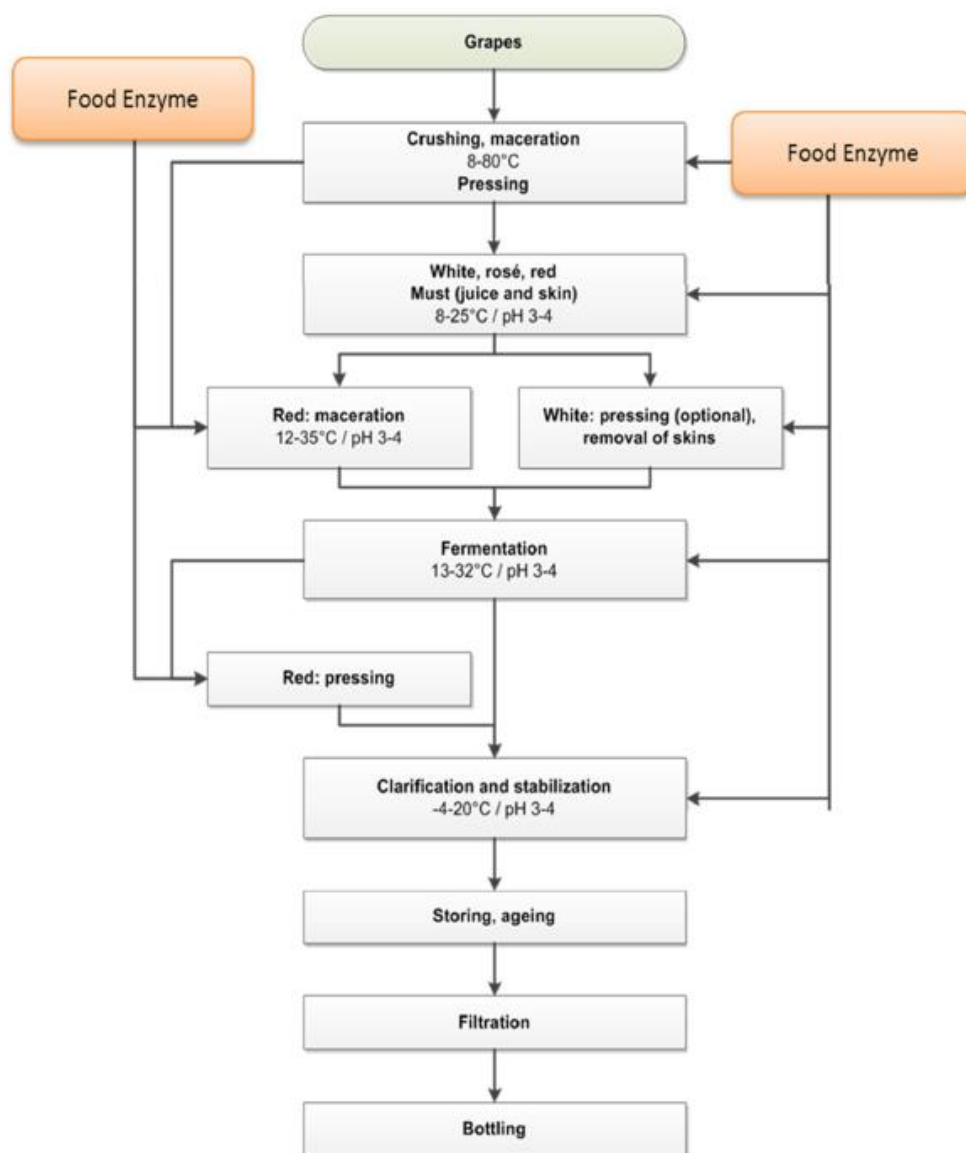
BREWING



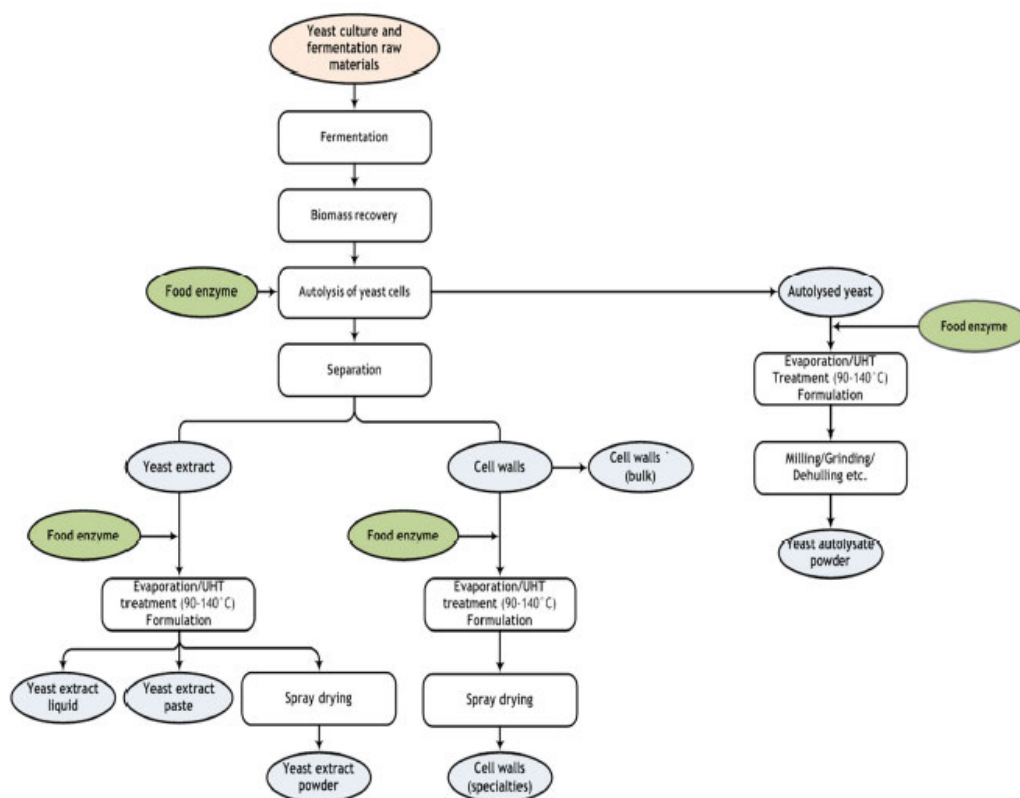
FRUIT AND VEGETABLE PROCESSING:



WINE MAKING



YEAST PROCESSING



2.8 Use Levels

Commercial food enzyme preparations are typically utilized in accordance with the Quantum Satis (QS) principle, meaning they are applied at a level no higher than necessary to achieve the desired enzymatic reaction, in accordance with GMP standards. The specific amount of enzyme activity added to the raw material will be established by the respective food manufacturer on a case by case basis, considering the intended effects and process conditions. Therefore, the enzyme manufacturer can only provide a recommended dosage range, which serves as a starting point for individual food producers to optimize their processes and determine the precise amount of enzyme needed to achieve their intended effect. Therefore, from a technological standpoint, there are no set "normal" or "maximum" usage levels, and Sumizyme PEG is used following the QS principle. A food producer using significantly higher doses than necessary would incur excessive costs and encounter adverse technological effects.

Microbial food enzymes contain not only the enzyme protein but also other substances derived from the fermentation medium and production microorganism. These constituents include organic materials such as proteins, peptides, amino acids, carbohydrates, lipids, as well as organic salts. As defined by JECFA (FAO/WHO, 2006), the percentages of these organic materials are combined and expressed as Total Organic Solids (TOS).

The TOS value is an internationally accepted method to describe the chemical composition of commercial food enzymes. The ratio between the enzyme activity and TOS is an indication of the relative purity of the enzyme. Whereas the dosage of a food enzyme depends on the enzyme activity present in the final food enzyme preparation, the dosage on basis of TOS is more relevant from a safety point of view. Therefore, the use levels are expressed in TOS.

The recommended use level ranges for the food enzyme in each of its applications are provided in Table 5, respectively.

Table 5. Recommended Use Levels of Sumizyme PEG

Application	Raw material (RM)	Recommended use levels (mg TOS/kg RM)	Maximal recommended use levels (mg TOS/kg RM)
Brewing	Cereals	0.1 - 2.0	2.0
Fruit and vegetable processing	Fruit and vegetables	0.2 - 2.0	2.0
Wine making	Grapes	0.1 - 11	11
Yeast processing	Yeast culture, yeast extract, cell walls, autolysed yeast	10 - 50	50

3 Part 3 – § 170.235 Dietary Exposure

As is outlined in Section 2.9, Sumizyme PEG may be used in the manufacture of a wide variety of foods, food ingredients and beverages.

Due to the diverse range of applications, the most suitable approach for estimating human consumption of food enzymes is the Budget Method (Hansen, 1966; Douglass *et al.*, 1997). This method allows for the calculation of a Theoretical Maximum Daily Intake (TMDI) based on conservative assumptions regarding physiological energy requirements from food as well as the energy density of food, rather than data from food consumption surveys. Originally developed for determining limits on food additive use, the Budget Method is known to provide conservative estimates of daily intake. It is based on assumed consumption levels (See Table 6 below) of key food items and beverages (with lower consumption levels assumed for less significant food items, such as snacks)

Table 6. Budget Method Assumed Daily Consumption of Food and Beverages

Average consumption over the course of a lifetime/kg body weight/day	Total solid food	Total non-milk beverages	Processed food (50% of total solid food)	Soft drinks (25% of total beverages)
	(kg)	(l)	(kg)	(l)
	0.025	0.1	0.0125	0.025

In addition to the assumptions from the Budget Method, it is assumed that wine and beer is consumed in the same amount as soft drinks. In Section 2.9, the recommended use levels of Sumizyme PEG are given, based on the raw materials used in the various food processes. The maximum use levels are selected for the calculation of the TMDI. In addition, the calculation considers the amount of food or beverage obtained per kg raw material, and the assumption is made that all of the TOS will end up in the final product. Based on the maximum recommended use levels, the maximum amount of Sumizyme PEG in final food products are provided in Table 7 below.

Table 7. Maximum Levels of Sumizyme PEG in Final Food Products

Application		Raw material (RM)	Maximal recommended use level (mg TOS/kg RM)	Example Final food (FF)	Ratio RM/FF	Maximal level in FF (mg TOS/kg food)
Beverages	Fruit and vegetable processing	Fruits and vegetables	2	Juice	1,3	2.6
	Wine making	Grapes	11	Wine	1,6	17.6
	Brewing	Cereals	2	Beer	0,17	0,34
Solid food	Yeast extract processing	Yeast culture, yeast extract, cell walls, autolysed yeast	50	Savoury snacks, flavouring, wine production, soups and bouillons, food supplements	0,02	1

Assumptions behind ratios of raw material to final food:

Brewing and cereal drinks

- Brewing and cereal drinks contribute to the category of liquid foods.

- Various types of grist (such as malt, barley, wheat, sorghum, and maize) are utilized as raw materials in the brewing and cereal drink processes. Yields vary depending on the type of grist, process used and the type of drink produced.
- Beer production involves a range of RM/FF from 14-28 kg of grist per 100 L of beer, with approximately 80 to 90% of all beers falling within the range of an RM/FF ratio of 14-20 kg of grist per 100 L of beer. The same RM/FF ratio applies to cereal beverage.
- The calculation of dietary exposure assumes a yield of 100 L of drink per 17 kg of cereal, corresponding to an RM/FF ratio of 0.17 kg of grist per liter of beer or cereal beverage.

Fruit and vegetable processing

- The main and most consumed food application in fruit and vegetable processing is fruit juice, which falls in the category of liquid foods.
- Raw materials used can be various fruits and vegetables and the yields will vary depending on the type of raw material used. Typically 0.75-0.9 L of juice is produced per kg of fruit, hence the range for RM/FF will be 1.1-1.3 kg fruit per L of fruit juice. The corresponding RM/FF ratio of 1.3 kg fruit per L of fruit juice will be used.

Wine making

- Wine adds to the class of liquid foods.
- Raw materials as well as production yield in wine making will vary depending on the type of wine produced. Raw material requirements for production of 100 L of wine are usually in the range of 130 – 160 kg grapes.
- The assumption for the calculation of dietary exposure is a yield of 100 L of wine per 160 kg of grapes, corresponding to a RM/FF ratio of 1.60 kg grapes per L of wine.

Yeast processing

- Food products from yeast processing fall in the category of solid foods. These processed yeast products are used predominantly as a savoury ingredient in a wide variety of foods or food ingredients, such as bouillon cubes or powder, sauces, gravy, ready to consume meals and processed cheese. According to input from EuraSYP (European Association for Specialty Yeast Products), the maximum amount of yeast product applied in the final food reaches the level of 2 %.

Based on the maximum values found in food and beverage (in the above cases wine and yeast extract), multiplied by the average consumption of food and beverage/kg body weight/day, the Total TMDI can be calculated. This calculation is provided in Table below:

Table 8. Calculation of the Total TMDI

TMDI in food (mg TOS/kg body weight/day)	TMDI in beverage (mg TOS/kg body weight/day)	Total TMDI (mg TOS/kg body weight/day)
$1 \times 0.0125 = 0.0125$	$17.6 \times 0.025 = 0.44$	0.4524

It is important to emphasize that the Total TMDI provided is based on conservative assumptions, resulting in a significantly inflated value for several reasons:

- It is assumed that all manufacturers of the aforementioned foods and beverages utilize the specific Sumizyme PEG enzyme;
- It is assumed that all manufacturers will apply the maximum use level per application;
- The calculation of TMDIs in both food and beverages only considers those items containing the highest theoretical amount of TOS. Consequently, items with lower theoretical amounts are not considered;
- It is assumed that the quantity of TOS remains constant throughout the food production process;
- It is assumed that the final food containing the calculated theoretical amount of TOS is ingested daily throughout one's lifetime;
- Assumptions on food and beverage intake among the general population are overestimates of the true average levels (Douglass *et al.*, 1997).

4 Part 4 - § 170.240 Self-Limiting Use Levels

This part is not applicable to this notified substance. Refer to Section 2.9 for additional details about use levels.

5 Part 5 - § 170.245 Experience Based on Common Use in Food before 1958

This part is not applicable to this notified substance.

6 Part 6 - § 170.250 GRAS Notice: Narrative

The information within this GRAS notice provides evidence that Sumizyme PEG is safe for use under the conditions described for use. This section will present the toxicological studies providing data supportive of this conclusion.

All available known information regarding the use of Sumizyme PEG has been reviewed and Erbslöh is not aware of any information that conflicts with the potential of this substance as being recognized as safe in the United States.

6.1 Safety of the production Strain

Pariza and Johnson (2001) Decision Tree

Guidelines for the safety assessment of microbial enzyme preparations have been published by Pariza and Johnson (2001). In this assessment, the toxigenic potential of the production organism is the basis for the safety of a given enzyme preparation. A decision tree is provided, which is followed by the questions and responses below. The outcome of this decision is that the Sumizyme PEG preparation is “ACCEPTED” as safe for its intended use.

1. Is the production strain genetically modified?

If yes, go to 2. If no, go to 6. NO

2. Is the production strain modified using rDNA techniques?

If yes, go to 3. If no, go to 5. NA

3. Issues related to the introduced DNA are addressed in 3a-3e.

3a. Do the expressed enzyme product(s) which are encoded by the introduced DNA have a history of safe use in food?

If yes, go to 3c. If no, go to 3b. NA.

3b. Is the NOAEL for the test article in appropriate short-term oral studies sufficiently high to ensure safety? If yes, go to 3c. If no, go to 12. NA

3c. Is the test article free of transferable antibiotic resistance gene DNA?

If yes, go to 3e. If no, go to 3d. NA

3d. Does (Do) the resistance gene(s) code for resistance to a drug substance used in treatment of disease agents in man or animals?

If yes, go to 12. If no, go to 3e. NA

3e. Is all other introduced DNA well characterized and free of attributes that would render it unsafe for constructing microorganisms to be used to produce food-grade products?

If yes, go to 4. If no, go to 12. NA

4. Is the introduced DNA randomly integrated into the chromosome?

If yes, go to 5. If no, go to 6. NA

5. Is the production strain sufficiently well characterized so that one may reasonably conclude that unintended pleiotropic effects, which may result in the synthesis of toxins or other unsafe metabolites will not arise due to the genetic modification method that was employed?

If yes, go to 6. If no, go to 7. NA

6. Is the production strain derived from a safe lineage, as previously demonstrated by repeated assessment *via* this evaluation procedure?

Many strains of this strain lineage exist, for which safety data are available, that can be or have been tested through the P&J Decision Tree evaluation scheme.

If yes, the test article is ACCEPTED. If no, go to 7. NO

7. Is the organism non-pathogenic?

If yes, go to 8. If no, go to 12. YES

8. Is the test article free of antibiotics?

If yes, go to 9. If no, go to 12. YES

9. Is the test article free of oral toxins known to be produced by other members of the same species?

If yes, go to 11. If no, go to 10. YES

10. Are the amounts of such toxins in the test article below levels of concern?

If yes, go to 11. If no, go to 12. NA

11. Is the NOAEL for the test article in appropriate oral studies sufficiently high to ensure safety?

Following 90-days of consecutive oral administration of the Sumizyme PEG food enzyme to rats at dose levels of up to 2,210 mg TOS/kg bodyweight/day, no toxicologically significant changes in the evaluated safety parameters were observed. The No Observed Adverse Effect Level (NOAEL) was, therefore, determined to be 2,210 mg TOS/kg bodyweight/day.

If yes, the test article is ACCEPTED. If no, go to 12. YES

12. An undesirable trait or substance may be present and the test article is not acceptable for food use. If the genetic potential for producing the undesirable trait or substance can be permanently inactivated or deleted, the test article may be passed through the decision tree again. NA

6.1.1 Pathogenicity and toxigenicity

Talaromyces versatilis does not appear on the list of pathogens in Annex III of Directive 2000/54/EC on the protection of workers from risks related to exposure to biological agent at work, as it is globally regarded as a safe microorganism:

- In the U.S., data within the GRAS notice for a cellulase enzyme from *Penicillium funiculosum* (the former name for *Talaromyces versatilis*) submitted in 2015 (GRN 584) to the FDA support that consumption of enzymes from this source organism are non-toxic.
- In Europe, *Talaromyces versatilis* is classified as a low-risk-class microorganism, as exemplified in the listing as Risk Group 1 in the microorganism classification lists of the German

Federal Institute for Occupational Safety and Health (BAuA) (BauA, 2016). It is not mentioned on the list of pathogens in Belgium (Belgian Biosafety Server, 2008), or the list of pathogenic biological agents in France (French Republic, 2021).

- As a result *Talaromyces versatilis* can be used under the lowest containment level at large scale, according to the ‘Good Industrial Large-Scale Practice’ (GILSP) criteria, as defined by the The Organisation for Economic Co-operation and Development (OECD, 1992).

There is no literature evidence of mycotoxin-producing *Talaromyces versatilis* strains (Dijksterhuis and Samson, 2007; Mapari et al. 2009; Frisvad et al. 2004; Labuda & Tančinová 2006). The EFSA reported that *P. funiculosum*, in general, produces no known mycotoxins except for a single strain that was reported to produce secalonin acid, according to unpublished data (EFSA, 2007a).

Furthermore, enzyme preparations with endo-1,3(4)- β -glucanase and xylanase activities derived from *Talaromyces versatilis* have recently been approved for use in animal feed industries by EFSA (EFSA, 2013, 2014). Similarly, a phytase feed enzyme preparation from *Talaromyces versatilis* was also approved (EFSA, 2007b).

Conclusion

The source material for the food enzyme is *Talaromyces versatilis*. The organism has not been genetically modified to produce the enzyme. The species *Talaromyces versatilis* has been used for decades for the production of food enzymes, under its former systematic name *Penicillium funiculosum*. *Talaromyces versatilis* has a long history of safe use in industrial scale enzyme production and can be considered as a safe production organism for enzymes for food and feed processing applications. In fermentation products derived from *Talaromyces versatilis*, secondary metabolites are not a safety concern. Therefore, *Talaromyces versatilis* can be considered generally safe as a production organism of food enzymes.

6.2 Safety of the Enzyme Preparation

6.2.1 Safety of β -glucanase enzymes

β -glucanase enzymes are commonly found in a variety of natural sources, including plants, bacteria, fungi and yeast (Nisha, 2022; Perrot *et al.* 2022). As such, there is a long history of exposure to these enzymes within the human diet, for which they are expected to be digested in the same manner as other dietary proteins. Furthermore, GRAS notifications on β -glucanase derived from various microbial sources, intended for similar uses as those for Sumizyme PEG, received letters with “no questions” from the FDA, such as GRN 149, 195, 479, 482, 535, 592 and 756. The FDA response to the first beta-glucanase GRAS notice, GRN 149, was in September of 2004, demonstrating its safe use for over 20 years in the U.S. Microbially derived β -glucanase enzymes have also been authorized for use in food production by other regulatory agencies around the globe, including Health Canada, JECFA, Food Standards Australia New Zealand (FSANZ) as well as EFSA, which recently authorized the use of the same enzyme that this the subject of this GRAS notice; endo-1,3(4)- β -glucanase produced from *Talaromyces*

versatilis strain PF8 (JECFA, 1995; EFSA, 2019, 2023, 2024; FSANZ, 2023; Health Canada, 2024).

The publicly available information that verifies the safety of other microbially derived β -glucanases is supportive of the safety of this β -glucanase, as the lack of toxicity of this class of enzymes is well established. While there is some natural variation in amino acid sequences and secondary/tertiary structure among enzymes of the same class, there is no basis that these differences may be a cause for toxicological concern, especially if the safety of the production organism has been demonstrated, since the enzymes share the same activity/function,

6.2.2 Information on Side Activities of the Enzyme Which May Cause Adverse Effects

As far as Erbslöh is aware, the β -glucanase described in this dossier does not possess any enzymatic side activities which might cause adverse effects. Microbial food enzymes are concentrates which typically contain minor amounts of other enzymatic activities (side activities) naturally produced by the microorganism. These microorganisms may produce numerous enzymes involved in the digestion of proteins, lipids and carbohydrates. Many of these hydrolase enzymes are produced endogenously in human within the gut microbiota, and are widely sold as digestive aids, in both foods and dietary supplements. These enzymes in digestive aids are sold at levels of exposure magnitudes higher than the negligible exposure from food enzymes used as processing aids in food manufacturing, are therefore enzymatic side activities are not relevant from a safety point of view.

6.2.3 Allergenicity

Since they are proteins, enzymes could theoretically have the potential to cause allergic responses. However, to address allergenicity by consumption, the following should be considered:

- Bindslev-Jensen *et al.* (2006) evaluated the allergenic potential of enzymes in the publication: "*Investigation on possible allergenicity of 19 different commercial enzymes used in the food industry*". This study involved enzymes produced by both wild-type and genetically modified strains as well as wild-type enzymes and Protein Engineered variants. 400 patients with a diagnosed allergy to inhalation allergens, food allergens, bee or wasp were included in the study. And it was concluded that in general, consumption of food enzymes are unlikely to pose a concern regarding food allergies.
- Previously, the Association of Manufacturers and Formulators of Enzyme Products (AMFEP) Working Group on Consumer Allergy Risk from Enzyme Residues in Food conducted a thorough examination on the allergenic potential of enzyme products (Dauvrin *et al.*, 1998). Overall, it was concluded that, unlike exposure *via* inhalation, there are no scientific evidence to suggest that the small quantities of enzymes present in food can sensitize or trigger allergy reactions in consumers.

- Enzymes have been consumed daily as digestive aids for decades, at levels much higher (up to 1 million times) in comparison to enzymes levels in food. Wuthrich (1996) published a list of enzymes used as digestive aids and concluded that they are not potent allergens by ingestion. Thus, there are no scientific indications that small amounts of enzymes in food can sensitize or induce allergic reactions in consumers.
- The majority of proteins do not cause food allergies, and the enzyme industry has not identified enzyme proteins used in food that are homologous to recognized food allergens (The only enzyme protein used in food that is recognized to have weak allergenic potential is egg lysozyme).
- During food processing, the food enzyme is used in small amounts, resulting in very small levels of enzyme protein in the final food products. Typically, a higher increases the risk of sensitization, while low levels in the final food are correlated with reduced risks (Goodman *et al.*, 2008).
- When proteins undergo denaturation, the enzyme molecule's tertiary structure is destroyed. Typically, these structural changes are linked with decreased antigenic reactivity in humans. In the majority of instances evaluated, exhibit significantly lower immunogenicity compared to their native counterparts (Valenta and Kraft, 2002; Valenta, 2002; Takai *et al.*, 1997; Takai *et al.*, 2000; Nakazawa *et al.*, 2005; Kikuchi *et al.*, 2006).
- Furthermore, residual enzyme proteins in the final food undergo digestion in the gastrointestinal tract, further reducing the risk of allergenicity of the enzyme. Although resistance to digestion is viewed as a potential allergenic risk factor, it is hypothesized that small protein fragments generated through digestion are less likely to induce allergies (FAO/WHO, 2001; Goodman *et al.*, 2008).
- Enzymes in food processing have a long history of safe use, showing no evidence of adverse effects or reactions. Additionally, a diverse array of enzyme classes and structures exist naturally in food, unlike most recognized food allergens, which are typically confined to a limited range of foods.

6.2.4 Allergenicity Search

To assess the potential risk of cross-reactivity between the β -glucanase enzyme and known allergens, sequence homology testing was conducted. This testing involved an 80-amino acid sliding window search and conventional FASTA alignment, with an overall homology threshold of 35%, as recommended in the most recent literature (FAO/WHO, 2001; Ladics *et al.* 2007; Goodman *et al.* 2008).

“AllergenOnline” database (<http://www.allergenonline.org>) was used to conduct the sequence homology comparison analysis.

This search demonstrated sections throughout the peptide sequence with greater than 35% identity to Asp f 7 from *Aspergillus fumigatus*, Der f 23 from *Dermatophagoides farina* and villin 2 from *Nicotiana tabacum*. The maximum sequence identity was 42.40%, to Asp f 7.

FASTA alignments of the sequences above with recognized allergens also exhibited matches to the Asp f 7 and Der f 23 (using E-value <0.1 as the cut-off) with 24.8 % and 32.4% identities respectively (0.00012 and 0.011 E-scores respectively) when using the complete sequence search function.

However, Asp f 7 and Der f 23 are not recognized as food allergens by the World Health organization and International Union of Immunological Societies (WHO/IUIS) Allergen Nomenclature Sub-committee.³

Although warned against in Codex (2009), it was highlighted by Herman *et al.* (2009) and expanded further by Ladics *et al.* (2011) along with information from AllergenOnline.org, that a short sequence of amino acids matches may not necessarily indicate cross-reactivity and could be overlooked by the more conservative 80 amino acid match ($\geq 35\%$). However, some regulatory bodies still require the search for isolated identity matches of 8 contiguous amino acids. When performed with the amino acid sequence for Sumizyme PEG, this search yielded no sequence matches with recognized allergens.

As for all enzyme products, the MSDS for Sumizyme PEG would include a precautionary statement indicating that inhalation of enzyme mist/dust could lead to allergic respiratory reactions, including asthma, particularly in susceptible individuals upon repeated exposure.

In summary, based on sequence homology alone, Sumizyme PEG is unlikely to present a risk of food allergenicity. Considering the outcomes of the bioinformatics analysis to estimate potential allergenicity based on similarity to known allergens, and taking into consideration the latest scientific guidance on interpreting this data, along with the understanding that the enzyme is usually denatured during food processing and any remaining enzyme in the final food undergoes digestion in the gastrointestinal tract, it is improbable that the β -glucanase produced by *Talaromyces versatilis* under assessment will trigger allergic reactions upon consumption of food containing residues of these enzymes.

6.2.5 Toxicological testing

In addition to the Pariza and Johnson assessment which concluded that the enzyme preparation can be accepted as safe without additional toxicity testing, the safety of the enzyme preparation was further established through the unpublished toxicological testing provided below.

An Ames test, Chromosomal Aberration study, Micronucleus test and 90-days oral toxicity study were performed in compliance with the principles of Good Laboratory Practice (GLP)

³ <https://www.allergen.org/viewallergen.php?aid=117> and <https://www.allergen.org/viewallergen.php?aid=934>

on a batch of the Sumizyme PEG food enzyme (Batch 110607T) with 22.1% TOS and enzyme activity of 1,070 U/mL. This percentage of TOS is slightly higher in the test substance than in commercially produced samples.

Bacterial Reverse Mutation Test with Sumizyme PEG (Experiment No. E376 (365-170))

The aim of this *in vitro* study was to detect the possible presence of compounds or contaminants in the test substance (batch 110607T) capable to induce a reverse mutation in bacteria *Salmonella* Typhimurium. This test enables the detection of base pair substitution and frameshift mutagens. The study was performed according to OECD guideline No. 471.

The mutagenicity of Sumizyme PEG was tested by a bacterial reverse mutation test using four histidine-auxotrophic strains of *Salmonella typhimurium* (TA100, TA98, TA1535 and TA1537) and one tryptophan-auxotrophic strain of *Escherichia coli* (WP2uvrA) in the absence (-S9 method) and presence (+S9 method) of the metabolic activation system. The preliminary dose-finding study was performed at the dose range of 0.086 to 22.1 mg TOS/plate (0.440 to 107 U/plate) for all strains. The number of revertant colonies at the highest dose increased- more than 2 times that of the negative control in strains TA100 in both the -S9 and +S9 methods, and TA1537 in the -S9 method at the second highest dose only (10.7 U/plate).

In the dose-finding study, a positive result was observed in strains TA100 in the -S9 method, and TA100, TA1535 and TA98 in the +S9 method. For the strains in which negative results were observed in the dose-finding study, the main study was conducted at the dose range of 0.69 to 22.1 mg TOS/plate (3.34 to 107 U/plate). Subsequently, a negative result was observed in strains WP2uvrA and TA1537 for both methods and TA1535 in the -S9 method, demonstrating a reproduction of the result of the dose-finding study. In contrast, a positive result was observed for strain TA98 in the -S9 method. For the strains TA1537 in the -S9 method and TA1535 and TA98 in the +S9 method, the result of the preliminary study was not reproduced in the dose-finding study. Therefore, the results of the preliminary study were deemed to be incidental due to the evaluation of only 1 plate.

In the preliminary study, dose-finding study and main study using the official OECD method, Sumizyme PEG accelerated the growth of background bacteria at higher doses in each strain. It has been reported that a similar phenomenon is often observed when using a substance containing free amino acids (Thompson *et al.* 2005; Khandoudi *et al.* 2009; Thybaud, *et al.* 2016). Thus, it seemed that the positive responses observed in the preliminary study and dose-finding study were caused by free amino acids in Sumizyme PEG, such as histidine. Thus, the bacterial reverse mutation test main study was conducted using a modified Induced Mutation Frequency (IMF) Test method in the dose range of 0.69 to 22.1 mg TOS/plate (3.34 to 107 U/plate). In the modified IMF method, bacteria are washed with a sodium-phosphate buffer after treatment with the test substance to exclude the effects of free amino acids. An additional confirmative study was conducted in the dose range of 1.38 to 22.1 mg TOS/plate (6.69 to 107 U/plate) to determine the reproducibility of the results in the bacterial reverse mutation test using the modified IMF method. Negative results were observed in all strains (TA100 and TA98 in

both methods and TA1535 in the +S9 method), and the reproducibility of the result was confirmed.

Therefore, it was concluded that Sumizyme PEG did not induce any gene mutation in bacteria under the conditions in this study. It was determined that the increases in the number of revertant colonies observed at the higher doses in strains TA100 and TA98 ($\pm$ S9 methods) and TA1535 (+S9 method) were caused by free amino acids such as histidine, in the test substance.

Chromosome aberration test with Sumizyme PEG in cultured mammalian cells - Experiment no. E377 (365-171)

The aim of this *in vitro* study was to detect the potential presence of compounds or contaminants in the test substance (batch 110607T) capable of inducing chromosomal breakage (clastogenesis) in cultured human lymphocytes. This test enables the detection of any chromosomal and chromatid structural aberration in the cells blocked at the metaphase stage. The study was performed according to the OECD guideline for the testing of chemicals No. 473.

This *in vitro* chromosome aberration test was conducting using Chinese hamster lung fibroblast cell lines (CHL/IU). The CHL cells were treated with the Sumizyme PEG or control substances under the following conditions: 1) Short-term treatment for 6 hours in the absence of S9 mix followed by an 18-hour incubation period in fresh media (-S9 assay), 2) Short-term treatment for 6 hours in the presence of S9 mix followed by an 18-hour incubation period in fresh media (+S9 assay), and 3) Continuous treatment for 24 hours in the absence of S9 mix (24h assay).

Physiological saline was used as a negative control, while mitomycin C (MMC) and cyclophosphamide (CP) were used as positive controls. The assay was conducted at 4 concentrations from 2.77 to 22.1 mg TOS/mL (13.4 to 107 U/mL as BC- β -glucanase activity, a dilution factor of 2) in the -S9, +S9 and 24h assays prior to microscopic examinations of cells for chromosome aberrations.

The incidences of structural chromosome aberrations in the Sumizyme PEG treated groups increased with statistically significant differences ($p \leq 0.025$) compared to the negative control, in only the highest dose group (22.1 mg TOS/mL; 107 U/mL) in the -S9 and 24h assays. In the +S9 assay, the incidences of structural chromosome aberrations were comparable to those in the negative controls at all concentrations. Furthermore, the incidences of polyploidy cells were comparable to those in the negative controls at all concentrations in all assays. The relative cell growth rates were 50% less at 22.1 mg TOS/mL (107 U/mL) in the 24h assay, although cell growth inhibition was not observed in the -S9 or +S9 assay.

An additional study was conducted to confirm the reproducibility or the dose-dependency in the 24h assays. Microscopic examinations of the cells were performed at 3 dose levels of 10.82, 15.47 and 22.1 mg TOS/mL (52.4, 74.9 and 107 U/mL) in the S9 assay, and at 4 dose levels of 7.58, 10.82, 15.47 and 22.1 mg TOS/mL (36.7, 52.4, 74.9 and 107 U/mL) in the 24 h assay. The incidences of structural chromosome aberrations in the Sumizyme PEG-treated groups were observed to be increased dose-dependently, and statistically significant differences ($p \leq 0.025$)

compared to the negative control were observed at dose levels ≥ 15.47 mg TOS/mL (74.9 U/mL) in the -S9 and 24h assays. The incidences of polyploidy cells in the treatment groups were comparable to the negative controls at all concentrations in both assays. The relative cell growth rates were 50% or less at 22.1 mg TOS/mL (107 U/mL) in the 24h assay, although cell growth inhibition was not observed in the -S9 assay.

It was concluded that Sumizyme PEG induced structural, but not numerical chromosome aberrations in cultured mammalian cells under the conditions of this study. However, it is known that the *in vitro* test for chromosome aberration is susceptible for false-positive results (Honda *et al.* 2018). Subsequently the food enzyme was further tested for genotoxicity in an *in vivo* micronucleus test, which tests for both structural and numerical chromosomal aberrations (EFSA, 2011; FDA, 2012). The *in vivo* micronucleus test with Sumizyme PEG is described below.

Micronucleus Assay of Sumizyme PEG in Rats Experiment No. E693 (365-182)

A micronucleus assay using Crl:CD (SD) male rats was conducted according to the OECD guideline for the testing of chemicals No. 474 to assess the potential of test substance Sumizyme PEG to induce micronucleated erythrocytes *in vivo*.

A Single Oral Dose Toxicity Study of Sumizyme PEG in Rats [Exp. No. E378 (365-172)] and 2-Week Repeated Oral Dose Toxicity Study of Sumizyme PEG in Rats [Exp. No. E379 (365-173)] were also conducted in which in which no adverse effects were observed.

Thus, the maximum dosage in these studies, a 10 mL/kg of a stock solution with 2.21 g TOS/kg (BC- β -glucanase activity: 10,700 U/kg), was selected as the high dose, and the half (1.105 g TOS/kg; 5,350 U/kg) and the quarter (0.5525 g TOS/kg; 2,680 U/kg) doses were selected for the micronucleus assay. Sumizyme PEG was orally administered to 6 rats/dose once a day, for 2 consecutive days. On 5 rats per dose, a microscopic examination was conducted to determine the frequency of micro-nucleated immature erythrocytes (MNIE) and the ratio of immature erythrocytes (IE) to analyzed erythrocytes at 24 hours after the final. No statistically significant increase in the frequency of MNIE was observed in any of the groups treated with Sumizyme PEG, compared with the negative control group. No significant difference in the ratio of IE was observed in any of the treated groups. In contrast, a statistically significant increase ($p \leq 0.025$) in the frequency of MNIE was observed in the positive control group, in which the animals were treated intravenously with a dose of 2 mg/kg mitomycin C.

Therefore, it was concluded that Sumizyme PEG did not induce micronucleated erythrocytes in rat bone marrow cells under the conditions in this study.

Given the initial positive results from the *in vitro* chromosomal aberration test, an *in vivo* micronucleus assay was conducted to further investigate Sumizyme PEG's genotoxicity under physiologically relevant conditions. While *in vitro* tests play a key role in preliminary genotoxicity assessments, they often yield false positives due to intrinsic limitations. *In vitro* systems, particularly those using mammalian cell lines, may not reliably reflect human metabolism, tissue distribution, or systemic interactions, all of which are important factors in

evaluating genotoxic potential (EFSA, 2011; FDA, 2012). The inability of *in vitro* models to replicate these complex processes is one reason *in vivo* assays are preferred for follow-up studies, as they provide a more accurate assessment of potential genotoxic effects in a living organism.

In genotoxicity testing, commonly used cell lines have several limitations that may contribute to a high rate of false positives. As noted in Kirkland *et al.* (2007), these cell lines often lack endogenous metabolic activity and rely on external metabolic systems. Furthermore, many of these cell lines have compromised p53 functions and altered DNA repair capabilities, which can lead to exaggerated responses to DNA damage under test conditions. High concentrations of test substances required by many *in vitro* protocols can also contribute to false-positive outcomes by inducing cytotoxicity rather than genuine genotoxicity (Kirkland *et al.* 2007).

This issue is further emphasized in regulatory guidance, which suggests careful interpretation of *in vitro* genotoxicity data, especially when tests yield positive results that are not corroborated by *in vivo* evidence (FDA, 2012). According to both EFSA (2011) and FDA (2012) guidelines, *in vivo* assays like the micronucleus test are valuable follow-ups because they address the limitations of *in vitro* assays and help distinguish between biologically relevant genotoxic effects and artifacts of the test system. Therefore, utilizing *in vivo* assays after an initial *in vitro* positive result aligns with regulatory best practices and ensures that findings are robust, reducing the likelihood of basing safety assessments on artifacts unique to *in vitro* conditions.

Overall, even though Sumizyme PEG was reported to induce structural chromosome aberrations, (but not numerical chromosome aberrations (polyploidy)) in the *in vitro* chromosomal aberration test, Sumizyme PEG was negative in the reverse mutation study in bacteria, demonstrating no direct effect on DNA, and no clastogenic potential in the *in vivo* micronucleus assay. Thus, it can be concluded that there is no expected genotoxicity potential resulting from exposure to Sumizyme PEG.

90-Day Repeated Oral Dose Toxicity Study of Sumizyme PEG in Rats - Experiment NO. E380 (365-174)

The aim of this study was to investigate the systemic toxicity potential of the test substance Sumizyme PEG, following 90-days of daily oral administration according to OECD guideline for the testing of Chemicals No. 408.

Sumizyme PEG was administered orally at a dose of 22.1, 221 or 2,210 mg TOS/kg bodyweight/day (107, 1,070 or 10,700 U/kg/day) to Crl:CD Sprague-Dawley rats (10/sex/group) for 90 consecutive days.

Throughout the administration period, the general condition of the animals was observed, and body weight as well as food consumption was measured. Urinalysis and ophthalmological examination were conducted just before the end of the administration period, while clinical laboratory examinations (hematology, blood coagulation, blood chemistry and serum protein

electrophoresis) and pathological examinations (organ weight measurement, macroscopic and histopathological examinations) were conducted after the administration period.

No mortality occurred during the study period, and no treatment related adverse effects on clinical signs, body weights, food consumption, ophthalmology, urinalysis, organ weights and histopathologic parameters were reported. While a few significant differences in some of these study parameters were observed relative to the control group, such as increased food consumption in female rats in the low dose group during days 85 to 90, the differences were slight, without any evidence of a dose-response relationship.

While a significant decrease in sodium and chloride levels of male rats in the high dose group were observed following the blood chemistry evaluation, the changes were slight, and the values were within normal historical ranges for this rat species. Furthermore, while a significant decrease in total protein and albumin was observed in the blood of females in the high dose group, the changes were slight, without evidence of dose-response relationship and there was no significant difference in the albumin/globulin (A/G) ratio.

Therefore, it was concluded that no test substance related effects on any of the parameters examined were observed, and that Sumizyme PEG was safe under the study conditions.

The Margin of Safety for human consumption was calculated by dividing the No Observed Adverse Effect Level (NOAEL) derived from the 90-day oral toxicity study by the Total Theoretical Maximal Daily Intake (TMDI). The calculations are performed on the basis of the maximum amounts of the food enzyme that could theoretically be carried over to final foods and drinks. In the present case, the values found for wine making and yeast extract were used to calculate the TMDI, which was found to be 0.4525 mg TOS/kg body weight/day. Consequently, the Margin of Safety is 4,884, even when calculated using an exaggerated TMDI, as described in Section 3.

In summary, the following information supports the safe use of the food enzyme:

- The enzyme was determined to be safe according to the decision tree by Pariza and Johnson (2001)
- No mutagenic activity was observed under the given test conditions of the Ames test
- The induction of structural chromosome aberrations, but not numerical chromosome aberrations (polyploidy) were observed under the given test conditions of the *in vitro* chromosomal aberration assay
- No induction of micronucleated erythrocytes were observed in the *in vivo* Micronucleus assay in rats, ruling out the concern for both structural and numerical chromosomal aberrations
- 90-day oral toxicity on rats: The No Observed Adverse Effect Level (NOAEL) was determined to be 2,210 mg TOS/kg bw/day, the highest dose tested in the study. The Margin of Safety was determined to be 4,884.

6.3 Safety Assessment and GRAS Determination

Based on the information provided throughout this notification, Erbslöh has determined that their β -glucanase enzyme preparation, Sumizyme PEG, is GRAS to be used *Quantum Satis* for its intended purpose in the processing of beer, wine, yeast, fruits and vegetables. Considering the high margin of safety, even when calculated using overestimated intake levels by the Budget method, it is not necessary to restrict the use of the enzyme in food processing. There is a long history of exposure to β -glucanases from natural sources within the human diet, as well as widespread use of these enzymes from various microbial sources, for use in food processing. β -glucanases have been approved by prominent regulatory agencies around the world, including the FDA, EFSA, Health Canada, JECFA and FSANZ, for similar uses as those intended for the subject of this GRAS notification. Sumizyme PEG is produced in accordance with GMP and HACCP principles, and the final product is free of heavy metal and microbiological contaminants. Furthermore, the safety of the production microorganism is well established and is not considered pathogenic. Toxicological studies conducted with the food enzyme provide corroborative evidence on the safe use of Sumizyme PEG for its intended uses, as the enzyme was demonstrated to lack genotoxic potential, and the NOAEL was determined to be 2,210 mg TOS/kg bw/day, which is 4,884 times higher than the TMDI of 0.4525 mg TOS/kg body weight/day.

7 Part 7 - §170.255 List of Supporting Data and Information in GRAS Dossier

Appendix 1. Certificates of Analyses

7.1 Literature References

BAuA. (2016). *Regelwerk - TRBA 460 einstuft von pilzen in risikogruppen*. Federal Institute for Occupational Safety and Health.

<https://www.baua.de/DE/Angebote/Regelwerk/TRBA/TRBA-460.html>

Belgian Biosafety Server. (2008). *Tools - Belgian classification for micro-organisms based on their biological risks*. Belgian Biosafety Server. <https://www.biosafety.be/content/tools-belgian-classification-micro-organisms-based-their-biological-risks>

Bindslev-Jensen C, Stahl Skov P, Roggen EL, Hvass P and Sidelmann Brinch D (2006). Investigation on possible allergenicity of 19 different commercial enzymes used in the food industry. *Food and Chemical Toxicology*, 44, 1909–1915

Bridge, P. D. (2013). *Talaromyces versatilis* Bridge & Buddie, sp. nov. *Index Fungorum*, 26(1).

Commission Regulation (EU) No 231/2012 of 9 March 2012 laying down specifications for food additives listed in Annexes II and III to Regulation (EC) No 1333/2008 of the European Parliament and of the Council. Official Journal L 83, p1, 22/03/2012.

<http://eurlex.europa.eu/LexUriServ/LexUriServ.do?uri=CONSLEG:2012R0231:20130206:EN:PDF>

Dauvrin T, Groot G, Maurer K-H, de Rijke D, Ryssov-Nielsen H, Simonse M and Sorensen TB (1998). Working group on consumer allergy risk from enzyme residues in food. An Amfep expert group evaluation study

Dijksterhuis J and Samson RA (editor) 2007. Food mycology: a multifaceted approach to fungi and food. CRC Press, Taylor & Francis Group, London.

Douglass JS, Barraj LM, Tennant DR, Long WR and Chaisson CF (1997). Evaluation of the Budget Method for screening food additive intakes. *Food Additives and Contaminants*, 14, 791-802

EFSA-Q-2005-281: Opinion of the Scientific Panel on Additives and Products or Substances used in Animal Feed and the Scientific Panel on Genetically Modified Organisms on the safety and efficacy of the enzyme preparation Rovabio™ PHY AP/LC (3-phytase) as feed additive for chickens for fattening, laying hens, piglets and pigs for fattening in accordance with Regulation (EC) No 1831/2003 The EFSA Journal (2007b) 471, 1-29. doi:10.2903/j.efsa.2007.471

EFSA-Q-2005-293: Introduction of a Qualified Presumption of Safety (QPS) approach for assessment of selected microorganisms referred to EFSA. The EFSA Journal (2007a) 587, 1-16. Annex 6: Assessment of filamentous fungi with respect to a Qualified Presumption of Safety. Available online at:
<http://www.efsa.europa.eu/de/scdocs/doc/587.pdf>

EFSA-Q-2010-01287: Scientific Opinion on Rovabio® Excel (endo-1,3(4)-beta-glucanase and endo-1,4-beta-xylanase) as a feed additive for chickens and turkeys for fattening, laying hens, piglets (weaned) and pigs for fattening, ducks, guinea fowls, quails, geese, pheasants and pigeons. EFSA Journal 2013; 11(7):3321.

EFSA-Q-2013-00750: Scientific Opinion on the safety and efficacy of Rovabio® Spiky (endo-1,4-beta-xylanase and endo-1,3(4)-beta-glucanase) as a feed additive for chickens for fattening, chickens reared for laying and other minor poultry species (for fattening and reared for laying). EFSA Journal 2014;12(7):3793 [20 pp.].
doi:10.2903/j.efsa.2014.3793

EFSA Panel on Food Contact Materials, Enzymes and Processing Aids (EFSA CEP Panel), Silano, V., Barat Baviera, J.M., Bolognesi, C., Brüschweiler, B.J., Cocconcelli, P.S., Crebelli, R., Gott, D.M., Grob, K., Lampi, E. and Mortensen, A., (2019). Safety evaluation of the food enzyme β -glucanase, xylanase and cellulase from *Mycothermus thermophiloides* (strain NZYM-ST). *EFSA Journal*, 17(3), p.e05631.

EFSA Panel on Food Contact Materials, Enzymes and Processing Aids (CEP), Lambré, C., Barat Baviera, J.M., Bolognesi, C., Cocconcelli, P.S., Crebelli, R., Gott, D.M., Grob, K., Lampi, E., Mengelers, M. and Mortensen, A., (2023). Safety evaluation of the food enzyme endo-1, 3 (4)- β -glucanase from the non-genetically modified *Rasamsonia composticola* strain 427-FS. *EFSA Journal*, 21(1), p.e07751.

EFSA Panel on Food Enzymes (FEZ), Zorn, H., Barat Baviera, J.M., Bolognesi, C., Catania, F., Gadermaier, G., Greiner, R., Mayo, B., Mortensen, A., Roos, Y.H. and Solano, M.L., (2024). Safety evaluation of the food enzyme endo-1, 3 (4)- β -glucanase from the non-genetically modified *Talaromyces versatilis* strain PF8. *EFSA Journal*, 22(10), p.e9033.

EFSA Scientific Committee, (2011). Scientific opinion on genotoxicity testing strategies applicable to food and feed safety assessment. *EFSA Journal*, 9(9), p.2379.

FAO/WHO (2001). Evaluation of allergenicity of genetically modified foods. Report of a Joint FAO/WHO Expert Consultation on Allergenicity of Foods Derived from Biotechnology.
http://www.who.int/foodsafety/publications/biotech/en/ec_jan2001.pdf

FAO/WHO (2006). General Specifications and Considerations for Enzyme Preparations Used in Food Processing. Compendium of food additive specifications, sixty-seventh meeting. FAO JECFA Monographs 3, 2006 (ISBN 92-5-105559-9), 63-67.
<ftp://ftp.fao.org/docrep/fao/009/a0675e/a0675e00.pdf>.

Food and Drug Administration. (2012). S2 (R1) genotoxicity testing and data interpretation for pharmaceuticals intended for human use. <https://www.fda.gov/media/71980/download>

French Republic. (2021). *Order of November 16, 2021 establishing the list of pathogenic biological agents*. Official Journal of the French Republic. <https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000044454319>

Frisvad JC, Smedsgaard J, Larsen TO & Samson RA (2004) Mycotoxins, drugs and other extrolites produced by species in *Penicillium* subgenus *Penicillium*. *Studies in Mycology* 49: 201-241.

[Food Standards Australia New Zealand](https://www.foodstandards.gov.au/australia-new-zealand) (FSANZ) (2023). Australia New Zealand Food Standards Code – Standard 1.3.3 – Processing aids. <https://www.legislation.gov.au/F2015L00402/latest/text>

Goodman RE, Vieths S, Sampson HA, Hill D, Ebisawa M, Taylor SL and van Ree R (2008). Allergenicity assessment of genetically modified crops--what makes sense? *Nat. Biotechnol.*, 26 (1), 73-78.

Hansen SC (1966). Acceptable daily intake of food additives and ceiling on levels of use. *Food Cosmet. Toxicol.* 4, 427–432.

Health Canada. (2024). List of Permitted Food Enzymes (Lists of Permitted Food Additives) <https://www.canada.ca/en/health-canada/services/food-nutrition/food-safety/food-additives/lists-permitted/5-enzymes.html>

Honda, H., Fujita, Y., Kasamatsu, T., Fuchs, A., Fautz, R., & Morita, O. (2018). Necessity for retrospective evaluation of past-positive chemicals in *in vitro* chromosomal aberration tests using recommended cytotoxicity indices. *Genes and Environment*, 40, 1-3.

Hong T-Y, Cheng C-W, Huang J-W and Meng M. (2002). Isolation and biochemical characterization of an endo-1,3-b-glucanase from *Streptomyces sioyaensis* containing a C-terminal family 6 carbohydrate-binding module that binds to 1,3-b-glucan. *Microbiology* 148:1151–1159.

Hua C, Yi H, Jiao L. (2011). Cloning and expression of the endo-1,3(4)- β -glucanase gene from *Paecilomyces* sp. FLH30 and characterization of the recombinant enzyme. *Biosci Biotechnol Biochem.* 75(9):1807-1812.

JECFA (Joint FAO/WHO Expert Committee on Food Additives), Evaluation of certain food additives and naturally occurring toxicants, 39th session, Geneva, 1995.

Khandoudi N, Porte P, Chtourou S, Nessler F, Marzin D, Le Curieux F. The presence of arginine may be a source of false positive results in the Ames test. *Mutat Res.* 2009 Sep-Oct;679(1-2):65-71.

Kikuchi T, Shibuya H and Jones JT. (2005). Molecular and biochemical characterization of an endo- β -1,3-glucanase from the pinewood nematode *Bursaphelenchus xylophilus* acquired by horizontal gene transfer from bacteria. *Biochem. J.* 389:117 – 125.

Kikuchi Y, Takai T, Kuhara T, Ota M, Kato T, Hatanaka H, Ichikawa S, Tokura T, Akiba H, Mitsuishi K, Ikeda S, Okumura K and Ogawa H (2006). Crucial commitment of proteolytic activity of a purified recombinant major house dust mite allergen Der p 1 to sensitization towards IgE and IgG responses. *J Immunol.*, 177, 1609-1617.

Kirkland, D., Pfuhler, S., Tweats, D., Aardema, M., Corvi, R., Darroudi, F., Elhajouji, A., Glatt, H., Hastwell, P., Hayashi, M., Kasper, P., Kirchner, S., Lynch, A., Marzin, D., Maurici, D., Meunier, J. R., Müller, L., Nohynek, G., Parry, J., Parry, E., White, P. (2007). How to reduce false positive results when undertaking in vitro genotoxicity testing and thus avoid unnecessary follow-up animal tests: Report of an ECVAM Workshop. *Mutation research*, 628(1), 31–55. <https://doi.org/10.1016/j.mrgentox.2006.11.008>

Labuda R & Tančinová D (2006). Fungi recovered from Slovakian poultry feed mixtures and their toxinogenicity. *Annals of Agricultural and Environmental Medicine* 13: 193-200.

Ladics, Gregory S.; Bannon, Gary A.; Silvanovich, Andre; Cressman, Robert F. (2007): Comparison of conventional FASTA identity searches with the 80 amino acid sliding window FASTA search for the elucidation of potential identities to known allergens. *In Mol. Nutr. Food Res.* 51 (8), pp. 985–998. DOI: 10.1002/mnfr.200600231.

Mapari SA, Meyer AS, Thrane U, Frisvad JC. (2009). Identification of potentially safe promising fungal cell factories for the production of polyketide natural food colorants using chemotaxonomic rationale. *Microb Cell Fact.* 2009 Apr 27;8:24. doi: 10.1186/1475-2859-8-24.

Nakazawa T, Takai T, Hatanaka H, Mizuuchi E, Nagamune T, Okumura K and Ogawa H (2005). Multiple-mutation at a potential ligand-binding region decreased allergenicity of a mite allergen Der f 2 without disrupting global structure. *FEBS Lett.*, 579, 1988-1994

Nisha, P. (2022). Beta-Glucanase: Diverse Bacterial Sources and its Applications. In *Microbial Beta Glucanases: Molecular Structure, Functions and Applications* (pp. 33-49). Singapore: Springer Nature Singapore.

Organisation for Economic Co-operation and Development (OECD). (1992). *Safety Considerations for Biotechnology 1992*. OECD. <https://www.oecd.org/sti/emerging-tech/2375496.pdf>

Pariza, M. W. and E. A. Johnson. (2001). Evaluating the Safety of Microbial Enzyme Preparations Used in Food Processing - Update for a New Century. *Regul. Toxicol. Pharmacol.* 33: 173-186.

Pariza MW and Foster EM. 1983. Determining the safety of enzymes used in food

processing. *J. Food Protection*, 46(5):453-468.

Perrot, T., Pauly, M., & Ramírez, V. (2022). Emerging roles of β -glucanases in plant development and adaptative responses. *Plants*, 11(9), 1119.

Samson RA, Yilmaz N, Houbraken J, Spierenburg H, Seifert KA, Peterson SW, Varga J, Frisvad JC. (2011). Phylogeny and nomenclature of the genus *Talaromyces* and taxa accommodated in *Penicillium* subgenus *Biverticillium*. *Stud Mycol.* 70(1):159-183.

Singh PD, Johnson JH, Aklonis CA & O'Sullivan J (1986) SQ 30,957, A new antibiotic produced by *Penicillium Funiculosum*. Taxonomy, fermentation, isolation, structure determination, synthesis and antibacterial activity. *The Journal of Antibiotics* 34(8): 1054-1058.

Takai T, Yokota T, Yasue M, Nishiyama C, Yuuki T, Mori A, Okudaira H and Okumura Y (1997). Engineering of the major house dust mite allergen Der f2 for allergen-specific immunotherapy. *Nat. Biotechnol.*, 15, 754 – 758.

Takai T, Ichikawa S, Yokota T, Hatanaka H, Inagaki F and Okumura Y (2000). Unlocking the allergenic structure of the major house dust mite allergen Der f 2 by elimination of key intramolecular interactions. *FEBS Lett.*, 484, 102 – 107.

Thompson, C., Morley, P., Kirkland, D., & Proudlock, R. (2005). Modified bacterial mutation test procedures for evaluation of peptides and amino acid-containing material. *Mutagenesis*, 20(5), 345–350. <https://doi.org/10.1093/mutage/gei045>

Thybaud, V., Kasper, P., Sobol, Z., Elhajouji, A., Fellows, M., Guerard, M., Lynch, A.M., Sutter, A. and Tanir, J.Y., 2016. Genotoxicity assessment of peptide/protein-related biotherapeutics: points to consider before testing. *Mutagenesis*, 31(4), pp.375-384.

Valenta R (2002). The future of antigen-specific immunotherapy of allergy. *Nat. Rev. Immunol.*, 2, 446 – 453.

Valenta R and Kraft D (2002). From allergen structure to new forms of allergen specific immunotherapy. *Curr. Opin. Immunol.*, 14, 718 – 727.

van Reenen-Hoekstra, E. S., Frisvad, J. C., Samson, R. A., & Stolk, A. C. (1990). The *Penicillium funiculosum* complex - Well defined species and problematic taxa. Modern concepts in *Penicillium* and *Aspergillus* classification, 173-192.

Wuthrich B (1996). Enzyme als ingestive Allergene. *Allergologie fur die Praxis*, 4, 74-91.

7.2 Appendix 1: Certificates of Analyses of the Food Enzyme and Food Enzyme Preparation

Certificates of Analyses of the Food Enzyme

Shin Nihon Chemical Co., Ltd.
19-10, Showa-cho, Anjo, Aichi 446-0063, Japan
Tel: +81-566-75-5555 Fax: +81-566-75-0010

Certificate of Analysis

This document certifies that this product complies with food Chemical Codex(4th ed.) recommended specifications for food grade enzyme preparations.

Product Name : Sumizyme PEG
Lot No. : 110402T
Manufacture date : April 22, 2011

BC-beta-glucanase Activity	827 u/g (4.0 Units/mg TOS)
Protein	10.8 %
Ash	0.8 %
Water	78.6 %
TOS	20.6 %
Arsenic	NMT 3 mg/kg
Heavy Metals (as Pb)	NMT 30 mg/kg
Lead	NMT 5 mg/kg
Total Viable Count	NMT 10 /g
Total Coliforms	NMT 30 /g
E.coli	Negative in 25 g
Salmonella	Negative in 25 g

Inspector

T. Konishi
Director
Quality Assurance

Shin Nihon Chemical Co., Ltd.

19-10, Showa-cho, Anjo, Aichi 446-0063, Japan

Tel: +81-566-75-5555 Fax: +81-566-75-0010

Certificate of Analysis

This document certifies that this product complies with food Chemical Codex(4th ed.) recommended specifications for food grade enzyme preparations.

Product Name : Sumizyme PEG

Lot No. : 120402T

Manufacture date : April 22, 2012

BC-beta-glucanase Activity	883 u/g (4.2 Units/mg TOS)
Protein	10.9 %
Ash	0.8 %
Water	78.2 %
TOS	21.0 %
Arsenic	NMT 3 mg/kg
Heavy Metals (as Pb)	NMT 30 mg/kg
Lead	NMT 5 mg/kg
Total Viable Count	NMT 10 /g
Total Coliforms	NMT 30 /g
E.coli	Negative in 25 g
Salmonella	Negative in 25 g

Inspector



T. Konishi
Director
Quality Assurance

Shin Nihon Chemical Co., Ltd.

19-10, Showa-cho, Anjo, Aichi 446-0063, Japan
Tel: +81-566-75-5555 Fax: +81-566-75-0010

Certificate of Analysis

This document certifies that this product complies with food Chemical Codex(4th ed.) recommended specifications for food grade enzyme preparations.

Product Name : Sumizyme PEG
Lot No. : 140303T
Manufacture date : March 23, 2014

BC-beta-glucanase Activity	830 u/g (3.9 Units/mg TOS)
Protein	11.1 %
Ash	0.9 %
Water	77.9 %
TOS	21.2 %
Arsenic	NMT 3 mg/kg
Heavy Metals (as Pb)	NMT 30 mg/kg
Lead	NMT 5 mg/kg
Total Viable Count	NMT 10 /g
Total Coliforms	NMT 30 /g
E.coli	Negative in 25 g
Salmonella	Negative in 25 g

Inspector



T. Konishi
Director
Quality Assurance

Example Certificate of Analysis of the Food Enzyme Preparation

E2200303

SHIN NIHON CHEMICAL CO., LTD.

19-10 Showa-cho, Anjo, Aichi 446-0063 Japan

Tel:+81-566-76-5171 Fax:+81-566-75-0010

Certificate of Analysis

This document certifies that this product complies with FOOD CHEMICALS CODEX recommended specifications for food grade enzyme preparations.

Product Name	:	Sumizyme PEG
Lot No.	:	221229-01
Analytical Date	:	January 13, 2023
Manufacture Date	:	December 29, 2022

Botrytis Glucanase Activity	300 u/g
Arsenic	NMT 3 mg/kg
Heavy Metals (as Pb)	NMT 30 mg/kg
Lead	NMT 5 mg/kg
Total Viable Count	NMT 10 /g
Total Coliforms	NMT 30 /g
E. coli	Negative in 25 g
Salmonella	Negative in 25 g
Yeast	NMT 10 /g
Mold	NMT 10 /g

Inspector

T. Takahama
Manager
Quality Assurance

DEPARTMENT OF HEALTH AND HUMAN SERVICES Food and Drug Administration GENERALLY RECOGNIZED AS SAFE (GRAS) NOTICE (Subpart E of Part 170)	Form Approved: OMB No. 0910-0342; Expiration Date: 07/31/2022 (See last page for OMB Statement)	
	FDA USE ONLY	
	GRN NUMBER	DATE OF RECEIPT
	ESTIMATED DAILY INTAKE	INTENDED USE FOR INTERNET
	NAME FOR INTERNET	
KEYWORDS		

Transmit completed form and attachments electronically via the Electronic Submission Gateway (*see Instructions*); OR Transmit completed form and attachments in paper format or on physical media to: Office of Food Additive Safety (*HFS-200*), Center for Food Safety and Applied Nutrition, Food and Drug Administration, 5001 Campus Drive, College Park, MD 20740-3835.

SECTION A – INTRODUCTORY INFORMATION ABOUT THE SUBMISSION

1. Type of Submission (<i>Check one</i>)	
☒ New	☐ Amendment to GRN No. _____ ☐ Supplement to GRN No. _____
2. ☒ All electronic files included in this submission have been checked and found to be virus free. (<i>Check box to verify</i>)	
3. Most recent presubmission meeting (<i>if any</i>) with FDA on the subject substance (<i>yyyy/mm/dd</i>): _____	
4. For Amendments or Supplements: Is your amendment or supplement submitted in response to a communication from FDA? (<i>Check one</i>)	
☐ Yes	If yes, enter the date of communication (<i>yyyy/mm/dd</i>): _____
☐ No	

SECTION B – INFORMATION ABOUT THE NOTIFIER

1a. Notifier	Name of Contact Person Dr. Jürgen Meinel		Position or Title Head of Quality and Regulatory Affairs	
	Organization (<i>if applicable</i>) Erbslöh Geisenheim GmbH			
	Mailing Address (<i>number and street</i>) Erbslöhstraße 1, Geisenheim, Germany			
City Geisenheim		State or Province Hessen	Zip Code/Postal Code 65366	Country Germany
Telephone Number +49 6722 7080		Fax Number	E-Mail Address juergen.meinl@erbsloeh.com	
1b. Agent or Attorney (<i>if applicable</i>)	Name of Contact Person Dylan Fronda		Position or Title Regulatory Scientific Specialist/Toxicologist	
	Organization (<i>if applicable</i>) dicentra			
	Mailing Address (<i>number and street</i>) 7 St Thomas St #603			
City Toronto		State or Province Ontario (ON)	Zip Code/Postal Code M5S 2B7	Country Canada
Telephone Number +1 866-647-3279		Fax Number	E-Mail Address dylan@dicentra.com	

SECTION C – GENERAL ADMINISTRATIVE INFORMATION

1. Name of notified substance, using an appropriately descriptive term

Endo-1,3(4)- β -glucanase enzyme preparation from *Talaromyces versatilis*

2. Submission Format: (Check appropriate box(es))

☒ Electronic Submission Gateway

☐ Electronic files on physical media

☐ Paper

If applicable give number and type of physical media

3. For paper submissions only:

Number of volumes _____

Total number of pages _____

4. Does this submission incorporate any information in CFSAN's files? (Check one)

☐ Yes (Proceed to Item 5)

☒ No (Proceed to Item 6)

5. The submission incorporates information from a previous submission to FDA as indicated below (Check all that apply)

☐ a) GRAS Notice No. GRN _____

☐ b) GRAS Affirmation Petition No. GRP _____

☐ c) Food Additive Petition No. FAP _____

☐ d) Food Master File No. FMF _____

☐ e) Other or Additional (describe or enter information as above) _____

6. Statutory basis for conclusions of GRAS status (Check one)

☒ Scientific procedures (21 CFR 170.30(a) and (b))

☐ Experience based on common use in food (21 CFR 170.30(a) and (c))

7. Does the submission (including information that you are incorporating) contain information that you view as trade secret or as confidential commercial or financial information? (see 21 CFR 170.225(c)(8) and 170.250(d) and (e))

☐ Yes (Proceed to Item 8)

☒ No (Proceed to Section D)

8. Have you designated information in your submission that you view as trade secret or as confidential commercial or financial information (Check all that apply)

☐ Yes, information is designated at the place where it occurs in the submission

☐ No

9. Have you attached a redacted copy of some or all of the submission? (Check one)

☐ Yes, a redacted copy of the complete submission

☐ Yes, a redacted copy of part(s) of the submission

☐ No

SECTION D – INTENDED USE

1. Describe the intended conditions of use of the notified substance, including the foods in which the substance will be used, the levels of use in such foods, and the purposes for which the substance will be used, including, when appropriate, a description of a subpopulation expected to consume the notified substance.

The endo-1,3(4)- β -glucanase enzyme preparation from *Talaromyces versatilis* is intended to be used in cereals for brewing and grapes for wine making, as well as in the processing of fruits, vegetables and yeast. The enzyme is used at levels no higher than the necessary dosage to achieve the desired enzymatic reaction and according to Good Manufacturing Practice. Consequently, there are no maximal use levels, only recommended dosages.

2. Does the intended use of the notified substance include any use in product(s) subject to regulation by the Food Safety and Inspection Service (FSIS) of the U.S. Department of Agriculture?

(Check one)

☐ Yes

☒ No

3. If your submission contains trade secrets, do you authorize FDA to provide this information to the Food Safety and Inspection Service of the U.S. Department of Agriculture?

(Check one)

☐ Yes

☐ No, you ask us to exclude trade secrets from the information FDA will send to FSIS.

SECTION E – PARTS 2 -7 OF YOUR GRAS NOTICE

(check list to help ensure your submission is complete – PART 1 is addressed in other sections of this form)

- ☒ PART 2 of a GRAS notice: Identity, method of manufacture, specifications, and physical or technical effect (170.230).
- ☒ PART 3 of a GRAS notice: Dietary exposure (170.235).
- ☒ PART 4 of a GRAS notice: Self-limiting levels of use (170.240).
- ☒ PART 5 of a GRAS notice: Experience based on common use in foods before 1958 (170.245).
- ☒ PART 6 of a GRAS notice: Narrative (170.250).
- ☒ PART 7 of a GRAS notice: List of supporting data and information in your GRAS notice (170.255)

Other Information

Did you include any other information that you want FDA to consider in evaluating your GRAS notice?

☒ Yes ☐ No

Did you include this other information in the list of attachments?

☒ Yes ☐ No

SECTION F – SIGNATURE AND CERTIFICATION STATEMENTS

1. The undersigned is informing FDA that Dr. Jürgen Meinel

(name of notifier)

has concluded that the intended use(s) of Endo-1,3(4)- β -glucanase enzyme preparation from *Talaromyces versatilis*

(name of notified substance)

described on this form, as discussed in the attached notice, is (are) not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act based on your conclusion that the substance is generally recognized as safe recognized as safe under the conditions of its intended use in accordance with § 170.30.

2. Dr. Jürgen Meinel agrees to make the data and information that are the basis for the conclusion of GRAS status available to FDA if FDA asks to see them; agrees to allow FDA to review and copy these data and information during customary business hours at the following location if FDA asks to do so; agrees to send these data and information to FDA if FDA asks to do so.

(name of notifier)

Erbslöhstraße 1, Geisenheim, Germany

(address of notifier or other location)

The notifying party certifies that this GRAS notice is a complete, representative, and balanced submission that includes unfavorable, as well as favorable information, pertinent to the evaluation of the safety and GRAS status of the use of the substance. The notifying party certifies that the information provided herein is accurate and complete to the best of his/her knowledge. Any knowing and willful misinterpretation is subject to criminal penalty pursuant to 18 U.S.C. 1001.

3. Signature of Responsible Official,
Agent, or Attorney

Printed Name and Title

Dylan Fronda, Regulatory Scientific Specialist/Toxicologist

Date (mm/dd/yyyy)

03/19/2024

SECTION G – LIST OF ATTACHMENTS

List your attached files or documents containing your submission, forms, amendments or supplements, and other pertinent information. Clearly identify the attachment with appropriate descriptive file names (or titles for paper documents), preferably as suggested in the guidance associated with this form. Number your attachments consecutively. When submitting paper documents, enter the inclusive page numbers of each portion of the document below.

Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	Erbsloh Geisenheim GmbH - GRAS Dossier Sumizyme PEG.pdf	Submission

OMB Statement: Public reporting burden for this collection of information is estimated to average 170 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden to: Department of Health and Human Services, Food and Drug Administration, Office of Chief Information Officer, PRASStaff@fda.hhs.gov. (Please do NOT return the form to this address). An agency may not conduct or sponsor, and a person is not required to respond to, a collection of information unless it displays a currently valid OMB control number.